THIRSTY'S CODEX

STANDARDS EDITION

VOLUME VIII / 08 OF 20

LAW, ASSURANCE, AND VALIDATION

Lifecycle Sections 219-257

NO PROOF, NO CLAIM

CONTINUOUS MASTER FOLIO / SOURCE-PRESERVING EDITION

Edition Statement

THIRSTY'S CODEX / STANDARDS EDITION / SOURCE-PRESERVING SET

This volume is one part of a continuous 20-volume edition. Canonical lifecycle section numbers remain stable across the set; no volume restarts or renumbers them.

The visual treatment has been rebuilt as a unified standards library: coordinated covers, volume architecture, source dividers, running headers, continuous master folios, local page numbers, and source-identity marks.

The source corpus remains authoritative over the presentation layer. Status terms are preserved exactly. CONTENT-COMPLETE does not silently become source-reconciled, hostile-reviewed, owner-accepted, ratified, or published.

Every registered source witness is identified by path, byte count, and SHA-256 digest. The complete source archive is preserved as an embedded attachment in Volume XX, while the two supplied foundational PDFs are preserved as attachments in Volume I.

CONTINUITY RULE

Read the master folio and canonical section sequence as one uninterrupted work. Each succeeding volume begins where the preceding volume ends.

Standards Edition Set Map

VOLUME I
Foundations and Governance
Doctrine, prompts, front matter, shared governance, roles, and output contracts

VOLUME II
Research, Strategy, and Risk
Lifecycle Sections 001-018

VOLUME III
Requirements through Hardware Architecture
Lifecycle Sections 019-060

VOLUME IV
Electrical Engineering through User Validation
Lifecycle Sections 061-106

VOLUME V
Software Systems through Smart Contracts
Lifecycle Sections 107-150

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Data, Intelligence, and Human Oversight
Lifecycle Sections 151-188

VOLUME VII
Cybersecurity through Privacy Engineering
Lifecycle Sections 189-218

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Law, Assurance, and Validation
Lifecycle Sections 219-257

VOLUME IX
Configuration through Delivery Infrastructure
Lifecycle Sections 258-300

VOLUME X
Production Operations and Command
Lifecycle Sections 301-332

VOLUME XI
Support, Enterprise, and Roadmaps
Lifecycle Sections 333-373

VOLUME XII
Maintenance, Decommissioning, and Retirement
Lifecycle Sections 374-406

VOLUME XIII
Reference Registers and Working Documentation
Indexes, cross-references, source registers, checkpoints, and operational guides

VOLUME XIV
Standards Source Library
Preserved standards, command editions, blueprints, and source artifacts

VOLUME XV
Canonical Authority and Assurance
Authority, constitutions, policies, schemas, provenance, evaluation, and verifica...

VOLUME XVI
Applied Corpus and V3Q
Agent-output corpus, V3Q implementation, fixtures, tests, and governed examples

VOLUME XVII
Training Views and Indexes
Continuous-pretraining, evaluation, retrieval, and training views with canonical ...

VOLUME XVIII
Agents and Owner Command Standards
Agents 101 and Owner's Cockpit source-standard facsimiles

VOLUME XIX
Static Analysis and Tooling Evidence
Codacy configuration, rulesets, analysis logs, and repository tooling

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219

SECTION 219 - LEGAL

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6,205 BYTES / PRESERVED SOURCE IDENTITY

SECTION 219 — Legal

STATUS: CONTENT-COMPLETE

PURPOSE

Identify, interpret, obtain advice on, and preserve the legal obligations, rights, authorities, liabilities, contracts, disputes, and jurisdictional constraints that affect the work.

REQUIRED OUTCOME

A legal-obligation map with jurisdictions, parties, claims, contracts, notices, approvals, privacy, safety, IP, records, regulatory, dispute, and escalation requirements, clearly marked as legal advice or unresolved counsel questions.

WHY THIS MATTERS

Legal risk is context-specific and can turn on facts, jurisdiction, timing, contract language, or procedural detail. A confident summary, generic compliance statement, or AI-generated interpretation is not a substitute for qualified counsel or accountable decision-making.

APPLICABILITY

Apply when law, regulation, contract, rights, litigation, investigation, licensing, intellectual property, employment, safety, privacy, public communication, or jurisdiction affects the scope or consequences.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when qualified review confirms no legal, contractual, rights, jurisdictional, or dispute effect for the exact scope, version, parties, and time. “Not legal work” is not sufficient.

PREREQUISITES AND DEPENDENCIES

Identify jurisdictions, entities, parties, activities, data, products, contracts, regulators, counsel, deadlines, records, and decision owners. Preserve legal holds and confidentiality requirements.

REQUIRED INPUTS

Objective, parties, jurisdictions, facts, contracts, policies, data, product or service, claims, notices, deadlines, approvals, prior advice, disputes, source identity, and unresolved legal questions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Qualified counsel or designated legal authority interprets law, privilege, litigation, disclosure, contract, and regulatory strategy. Business owners provide truthful facts, implement approved obligations, and accept non-legal operational tradeoffs only within authority.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize authorized legal sources, compare clauses, extract obligations, draft questions and issue registers, and summarize supplied advice with provenance. It must not claim to be counsel, create attorney-client privilege, infer jurisdiction, provide definitive legal advice, conceal adverse facts, or approve a legal position or disclosure.

APPLICABLE EXISTING AGENT ROLES

Legal Advisor, Researcher, Analyst, Governance Reviewer, Privacy Reviewer, Records Manager, Documentation Writer, Verifier, Critic, and Memory Curator. Legal advice remains with qualified human authority.

REQUIRED ACTIVITIES

Define facts and scope; identify jurisdictions and parties; inventory obligations and contracts; obtain counsel review; map duties to controls and owners; track deadlines and notices; preserve holds; record assumptions and conflicts; verify implementation; and escalate unresolved questions.

HUMAN-AI INTERACTION REQUIREMENTS

Label source, jurisdiction, date, authority, privilege or confidentiality, fact, interpretation, question, recommendation, and decision. Keep sensitive advice in approved channels. Require counsel or authorized human review for material legal conclusions, notices, claims, and disclosures.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent obligations, parties, jurisdictions, contracts, rights, deadlines, evidence, holds, approvals, and controls as versioned records with access boundaries. Prevent automated workflows from silently changing legal position or destroying held records.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show obligation, jurisdiction, source, counsel status, owner, deadline, hold, risk, evidence, unresolved question, and authorized assign, approve, restrict, notify, escalate, preserve, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Legal issue and obligation register; jurisdiction and party map; contract and notice inventory; counsel questions and advice record; hold and deadline record; control mapping; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, jurisdiction, version, party, fact, date, authority, method, reviewer, conclusion or question, and limitation. A legal summary, template, policy, or AI output is not proof of legal sufficiency.

VERIFICATION AND ACCEPTANCE TESTS

Verify source and jurisdiction, party and contract scope, deadline, privilege handling, hold, implementation, notice, approval, disclosure, and escalation. Obtain qualified review for material claims and retain the basis for acceptance.

FAILURE STATES AND ANTI-PATTERNS

AI presented as counsel, law copied without jurisdiction, stale advice, hidden facts, privilege in uncontrolled systems, contracts ignored as operational inputs, and compliance asserted from a checklist.

RECOVERY AND RECONSIDERATION

Pause affected action, preserve records, escalate to counsel, correct the factual record, notify authorized owners, and reassess deadlines and exposure. Reopen on new facts, jurisdiction, contract, law, regulator, dispute, or product change.

COMPLETION CRITERIA

Facts, jurisdictions, parties, obligations, sources, counsel status, owners, deadlines, controls, evidence, unresolved questions, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve source, advice status, privilege, holds, deadlines, decisions, notices, contracts, evidence, and open gaps. Hand off only through authorized channels to Compliance, Policy, Privacy, Records, IP, Licensing, Security, and accountable owners.

RELATED CODEX SECTIONS

004 Evidence and Truth; 018 Risk Management; 047 Privacy Architecture; 220 Regulatory Compliance; 221 Policy; 222 Ethics; 223 Intellectual Property; 224 Licensing; 225 Records Management; 402-406 Review and Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence requirements, and the integration roadmap. No jurisdiction, counsel, contract set, regulator, privilege rule, deadline, or legal acceptance authority is assumed by this draft.

220

SECTION 220 - REGULATORY COMPLIANCE

CODEX/03_SECTIONS/220_REGULATORY_COMPLIANCE.txt

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5,869 BYTES / PRESERVED SOURCE IDENTITY

SECTION 220 — Regulatory Compliance

STATUS: CONTENT-COMPLETE

PURPOSE

Translate applicable regulatory obligations into owned, testable, evidenced, and continuously reviewed controls without confusing compliance documentation with actual protection or lawful operation.

REQUIRED OUTCOME

A regulatory register mapping jurisdiction, authority, obligation, scope, control, owner, evidence, cadence, reporting, exception, remediation, and change trigger.

WHY THIS MATTERS

Regulation varies by jurisdiction, sector, activity, data, product, and time. A certification, policy, audit, or control description can be stale or incomplete and does not itself prove ongoing compliance.

APPLICABILITY

Apply when a regulator, statute, rule, license, order, standard adopted by contract, or public obligation affects the system, organization, product, data, or process.

NON-APPLICABILITY CONDITIONS

Record non-applicability only after qualified review of jurisdiction, entity, activity, data, population, product, and time. Preserve the reason, source, reviewer, date, and reopening condition.

PREREQUISITES AND DEPENDENCIES

Identify legal entities, jurisdictions, regulators, sector, data, contracts, products, controls, counsel, compliance owner, evidence systems, reporting deadlines, and audit or inspection authority.

REQUIRED INPUTS

Regulatory sources, scope, obligations, definitions, controls, policies, procedures, assets, owners, evidence, assessments, incidents, findings, reports, exceptions, and change notices.

HUMAN RESPONSIBILITIES AND AUTHORITY

Compliance and legal owners interpret obligations with qualified counsel, approve applicability, control mapping, reports, exceptions, and regulator communication. Control owners implement and evidence obligations. Humans accept residual risk and attest only to supported facts.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize authorized sources, extract obligations, compare control evidence, identify gaps, draft questions, and track changes. It must preserve jurisdiction, source, date, definition, evidence, and uncertainty. It must not declare compliance, provide legal advice, sign an attestation, conceal gaps, or communicate to a regulator without authorized human approval.

APPLICABLE EXISTING AGENT ROLES

Compliance Analyst, Legal Advisor, Researcher, Governance Reviewer, Auditor, Security Auditor, Privacy Reviewer, Analyst, Verifier, Documentation Writer, Critic, and Memory Curator.

REQUIRED ACTIVITIES

Identify scope and authorities; inventory obligations; map controls; assign owners and cadence; collect evidence; assess design and operation; track findings and exceptions; prepare reports; monitor regulatory change; test; obtain approval; and preserve continuity.

HUMAN-AI INTERACTION REQUIREMENTS

Show authority, jurisdiction, obligation, source date, control, evidence, owner, status, limitation, and required decision. Require qualified human review for interpretation, applicability, attestation, exception, report, and regulator communication.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent obligations, controls, evidence, owners, attestations, findings, deadlines, and source versions as traceable records. Preserve segregation of duties, evidence integrity, access restrictions, legal holds, and change impact across systems.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show obligation, authority, scope, control, owner, evidence freshness, status, finding, deadline, exception, and authorized assign, attest, remediate, escalate, report, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Regulatory applicability and obligation register; control map; evidence and cadence plan; assessment and findings; exception and remediation record; report and approval; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify authority, jurisdiction, source version, obligation, scope, control, environment, method, time, owner, result, and limitation. A policy, certificate, or audit report proves only its stated scope and period.

VERIFICATION AND ACCEPTANCE TESTS

Verify applicability, source currency, control implementation and operation, evidence integrity, sampling, segregation, exceptions, reporting, deadlines, and change monitoring. Obtain qualified acceptance for material conclusions.

FAILURE STATES AND ANTI-PATTERNS

Checklist compliance, stale register, unsupported attestation, control description treated as operation, one jurisdiction generalized to another, evidence manufactured for the audit, and exceptions without expiry or owner.

RECOVERY AND RECONSIDERATION

Mark unsupported claims, preserve evidence, notify compliance and legal owners, remediate or disclose through approved channels, and reassess scope and deadlines. Reopen on regulatory, product, data, entity, jurisdiction, contract, or control change.

COMPLETION CRITERIA

Authority, applicability, obligations, controls, owners, evidence, cadence, findings, exceptions, reports, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve source and obligation versions, control mappings, evidence, findings, attestations, reports, communications, holds, exceptions, and open gaps. Hand off to Legal, Policy, Audit, Security, Privacy, Records, Operations, and Governance.

RELATED CODEX SECTIONS

004 Evidence and Truth; 018 Risk Management; 219 Legal; 221 Policy; 225 Records Management; 226 Audit; 227-230 Internal Controls and Quality; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence requirements, and the integration roadmap. Exact jurisdictions, regulators, sector rules, reporting, attestation, counsel, and acceptance authority remain unresolved.

221

SECTION 221 - POLICY

CODEX/03_SECTIONS/221_POLICY.txt

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5,725 BYTES / PRESERVED SOURCE IDENTITY

SECTION 221 — Policy

STATUS: CONTENT-COMPLETE

PURPOSE

Create, approve, communicate, implement, monitor, interpret, review, and retire governing statements that direct behavior, authority, risk, rights, and controls.

REQUIRED OUTCOME

A policy lifecycle with purpose, scope, authority, definitions, obligations, exceptions, enforcement, implementation owners, communication, evidence, review, appeal, change, and retirement.

WHY THIS MATTERS

Policies can be ambiguous, contradictory, inaccessible, unenforced, or detached from actual work. A signed document is not proof that people understood it, systems implemented it, or exceptions were governed.

APPLICABILITY

Apply when a rule, standard, principle, commitment, restriction, or decision framework governs people, systems, data, products, operations, suppliers, or agents.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no governing behavior, authority, protected outcome, or control is affected and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Identify authority, affected parties, purpose, legal and regulatory sources, existing policies, processes, controls, enforcement, accessibility, communication, and review owner.

REQUIRED INPUTS

Objective, scope, authority, definitions, principles, requirements, exceptions, stakeholders, risks, evidence, implementation, communication, appeals, review cadence, and source conflicts.

HUMAN RESPONSIBILITIES AND AUTHORITY

Policy owners draft and maintain policy; authorized humans approve, interpret material ambiguity, grant exceptions, resolve conflicts, communicate changes, and retire obsolete rules. Affected people must have a route to question, appeal, or report harm.

AI RESPONSIBILITIES AND BOUNDARIES

AI may compare sources, identify contradictions, draft language, map policy to controls, and summarize feedback. It must preserve authority, source, version, scope, and uncertainty. It must not silently change meaning, create obligations, interpret law conclusively, approve, enforce, or hide dissent without human authority.

APPLICABLE EXISTING AGENT ROLES

Policy Analyst, Legal Advisor, Governance Reviewer, Researcher, Analyst, User Experience Reviewer, Accessibility Reviewer, Critic, Verifier, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define need and authority; inventory sources; draft and review; resolve conflicts; assess impact and accessibility; approve and publish; map to controls and procedures; train and communicate; monitor exceptions and outcomes; review and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, authority, version, scope, definition, interpretation, affected party, and uncertainty. Require human approval for policy, exception, enforcement, legal meaning, and material communication. Present changes and impacts clearly and accessibly.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent policy, version, authority, scope, definitions, controls, exceptions, decisions, acknowledgments, and evidence as linked records. Ensure enforcement systems reference an approved version and fail safely when policy is unavailable or conflicting.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show policy status, owner, authority, scope, version, effective date, controls, exceptions, acknowledgments, findings, evidence freshness, and authorized approve, communicate, interpret, exempt, review, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Policy charter and text; authority and source register; impact and accessibility review; control mapping; exception and appeal procedure; communication and training; evidence, decision, review, and continuity records.

EVIDENCE REQUIREMENTS

Identify policy, authority, version, scope, effective date, source, reviewer, control, implementation, communication, result, and limitation. Publication or acknowledgment proves neither comprehension nor operation.

VERIFICATION AND ACCEPTANCE TESTS

Verify authority, definitions, conflicts, scope, accessibility, implementation, communication, control behavior, exceptions, appeal, review, and retirement. Sample actual operations and user understanding rather than relying only on signatures.

FAILURE STATES AND ANTI-PATTERNS

Policy sprawl, contradictory rules, undefined terms, secret policy, inaccessible text, no owner, exceptions without expiry, and enforcement that differs from the approved policy.

RECOVERY AND RECONSIDERATION

Pause or narrow affected enforcement, preserve versions and decisions, notify owners, correct communication, remediate harm, and use the prior valid policy where authorized. Reopen on legal, operational, technological, user, or risk change.

COMPLETION CRITERIA

Purpose, authority, scope, sources, definitions, controls, exceptions, communication, evidence, review, appeal, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending authorized approval and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve policy versions, approvals, interpretations, communications, acknowledgments, exceptions, evidence, findings, and open gaps. Hand off to Legal, Compliance, Governance, Operations, Security, Privacy, Support, and Documentation.

RELATED CODEX SECTIONS

003 Authority and Delegation; 004 Evidence and Truth; 018 Risk Management; 219 Legal; 220 Regulatory Compliance; 222 Ethics; 225 Records Management; 226-230 Audit and Quality.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, authority and evidence requirements, and the integration roadmap. Exact policy authority, legal source, audience, enforcement, exception, accessibility, and acceptance requirements remain unresolved.

222

SECTION 222 - ETHICS

CODEX/03_SECTIONS/222_ETHICS.txt

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5,956 BYTES / PRESERVED SOURCE IDENTITY

SECTION 222 — Ethics

STATUS: CONTENT-COMPLETE

PURPOSE

Make value conflicts, foreseeable harms, affected parties, power differences, duties, tradeoffs, and accountability explicit in decisions that are not settled by technical or legal analysis alone.

REQUIRED OUTCOME

An ethics assessment with human purpose, affected parties, values, options, harms, benefits, uncertainty, dissent, safeguards, decision authority, monitoring, appeal, and reconsideration.

WHY THIS MATTERS

A technically feasible or legally permitted action can still be harmful, unfair, exploitative, deceptive, or inconsistent with the human purpose. Ethics work must not become a decorative approval after the decision is already fixed.

APPLICABILITY

Apply when choices affect people, rights, dignity, safety, opportunity, privacy, communities, environment, future users, or power relationships, especially when automation or AI influences outcomes.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when the change has no affected person, value conflict, material externality, or consequential choice and an approved exact baseline supports it. Technical simplicity is not a sufficient reason.

PREREQUISITES AND DEPENDENCIES

Use user research, risk, privacy, safety, security, legal, accessibility, policy, impact, and governance evidence. Identify affected and absent voices, decision owner, escalation, and review authority.

REQUIRED INPUTS

Objective, options, stakeholders, affected non-users, values, constraints, benefits, harms, uncertainty, evidence, alternatives, power differences, safeguards, monitoring, and unresolved dissent.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans identify values, listen to affected people, resolve moral tradeoffs, set unacceptable harm, approve safeguards, and accept or reject residual impact. No agent may transfer that responsibility to a score, majority, policy, or technical metric.

AI RESPONSIBILITIES AND BOUNDARIES

AI may surface affected parties, compare options, identify value conflicts, summarize supplied perspectives, and challenge assumptions. It must not claim moral authority, manufacture consensus, erase dissent, infer values from proxy data, or decide human worth, rights, harm, or acceptable tradeoff without accountable human deliberation.

APPLICABLE EXISTING AGENT ROLES

Ethics Reviewer, Researcher, User Experience Reviewer, Analyst, Critic, Red Team, Privacy Reviewer, Legal Advisor, Safety Reviewer, Governance Reviewer, Verifier, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Frame the decision; identify affected and missing voices; articulate values and conflicts; generate alternatives; assess harms and benefits; test assumptions; design safeguards and recourse; document dissent; obtain accountable decision; monitor outcomes; and set reopening triggers.

HUMAN-AI INTERACTION REQUIREMENTS

Show whose interests, evidence, values, uncertainty, assumptions, and tradeoffs are represented. Require human deliberation for material impact. Make participation accessible and do not present AI-generated balance as actual consultation.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent purpose, affected parties, values, harms, safeguards, decisions, dissent, monitoring, appeal, and evidence as durable records. Build in human override, transparency, privacy, accessibility, and recovery for systems that operationalize ethical commitments.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show decision, affected parties, values, options, harms, safeguards, dissent, owner, evidence, unresolved questions, and authorized pause, reject, revise, escalate, appeal, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Ethics charter; affected-party and value map; options and harm assessment; safeguard and recourse design; dissent record; decision, monitoring, review, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify claim, affected party, source, perspective, method, date, assumption, uncertainty, decision, safeguard, and limitation. A principle, survey, model output, or legal permission is not complete ethical evidence.

VERIFICATION AND ACCEPTANCE TESTS

Verify affected-party coverage, alternatives, harm analysis, safeguard operation, accessibility, recourse, dissent preservation, outcome monitoring, and decision authority. Revisit with new evidence and lived experience.

FAILURE STATES AND ANTI-PATTERNS

Ethics washing, token consultation, consensus theater, value substitution by metrics, absent affected parties, harm externalized to powerless people, and “legal means ethical” reasoning.

RECOVERY AND RECONSIDERATION

Pause or narrow harmful action, protect affected people, preserve evidence, seek missing perspectives, remedy where possible, and revisit the decision with accountable owners. Reopen on new harm, population, purpose, evidence, technology, or value change.

COMPLETION CRITERIA

Purpose, affected parties, values, options, harms, safeguards, authority, evidence, dissent, recourse, monitoring, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent challenge and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve assessment, perspectives, dissent, decision, safeguards, incidents, monitoring, evidence, exceptions, and open gaps. Hand off to Legal, Privacy, Safety, Trust and Safety, Policy, Governance, UX, Testing, and Operations.

RELATED CODEX SECTIONS

001 Research; 004 Evidence and Truth; 018 Risk Management; 019 Requirements; 047 Privacy Architecture; 217 Trust and Safety; 218 Privacy Engineering; 219 Legal; 221 Policy; 402-404 Review.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, human-AI interaction requirements, and the integration roadmap. Exact ethical framework, affected communities, participation, authority, remedies, jurisdiction, and acceptance evidence remain unresolved.

223

SECTION 223 - INTELLECTUAL PROPERTY

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5,970 BYTES / PRESERVED SOURCE IDENTITY

SECTION 223 — Intellectual Property

STATUS: CONTENT-COMPLETE

PURPOSE

Identify, protect, attribute, license, use, create, transfer, and retire intellectual-property interests and restrictions affecting the work.

REQUIRED OUTCOME

An IP register with works, inventions, marks, data, model or training material, owners, contributors, provenance, rights, restrictions, licenses, approvals, confidentiality, publication, enforcement, and evidence.

WHY THIS MATTERS

Ownership and permission can be unclear across employees, contractors, suppliers, open-source projects, generated material, data, and jurisdictions. A repository or citation is not proof of permission, ownership, originality, or compliant reuse.

APPLICABILITY

Apply when the work creates, uses, modifies, distributes, trains on, embeds, names, publishes, or derives value from potentially protected works, inventions, data, marks, or confidential material.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when qualified review confirms no protected or restricted material, contributor claim, license, confidentiality, publication, or transfer is affected.

PREREQUISITES AND DEPENDENCIES

Use Legal, Licensing, Records, Security, Privacy, Procurement, employment, supplier, and source-control baselines. Identify contributors, employers, licensors, repositories, jurisdictions, confidentiality, and approval authority.

REQUIRED INPUTS

Assets, authors, contributors, source and provenance, licenses, contracts, data, dependencies, marks, inventions, confidentiality, publication plan, notices, restrictions, and unresolved ownership questions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners and qualified counsel approve ownership, assignment, license, attribution, publication, disclosure, enforcement, and use of restricted material. Humans resolve contributor disputes and decide whether a source or generated artifact may be used.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory supplied sources, compare license text, track provenance, draft attribution and review questions, and identify possible conflicts. It must not assert originality, ownership, permission, authorship, or legal sufficiency; it must not ingest restricted material or publish, license, or disclose work without authorized human decision.

APPLICABLE EXISTING AGENT ROLES

Legal Advisor, IP Analyst, Researcher, Governance Reviewer, Documentation Writer, Security Auditor, Privacy Reviewer, Verifier, Critic, and Memory Curator.

REQUIRED ACTIVITIES

Inventory and classify IP; capture provenance and contributors; review contracts and licenses; map obligations; protect confidential material; obtain assignments or permissions; preserve notices and attribution; approve publication; monitor use; handle claims and retirement.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, contributor, license, jurisdiction, confidence, restriction, and unresolved question. Keep confidential or restricted material in approved systems. Require human or counsel review for ownership, permission, attribution, infringement, publication, and enforcement.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent IP assets, provenance, license, contributor, restrictions, approvals, notices, and distribution as linked versioned records. Enforce repository, build, model, data, and publication gates and prevent unauthorized material from entering generated or shipped artifacts.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show asset, owner, contributor, source, license, restrictions, approval, publication state, evidence freshness, and authorized review, restrict, correct, attribute, withdraw, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

IP and provenance register; contributor and ownership record; license and restriction map; confidentiality and publication review; attribution and notice package; claim, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, source, author, contributor, date, license or contract, jurisdiction, approval, version, method, result, and limitation. A URL, repository, generated output, or citation proves neither permission nor ownership.

VERIFICATION AND ACCEPTANCE TESTS

Verify provenance, contributor rights, license compatibility, attribution, notices, confidentiality, publication, build inclusion, data or model restrictions, and withdrawal paths. Obtain qualified review for material uncertainty.

FAILURE STATES AND ANTI-PATTERNS

Untracked copied code or data, license mixing, missing attribution, confidential material in public artifacts, authorship assumed from tool output, and publication before rights review.

RECOVERY AND RECONSIDERATION

Restrict distribution, preserve provenance and claim evidence, notify owners and counsel, replace or remove affected material, correct notices, and assess downstream copies. Reopen on new source, contributor, license, jurisdiction, publication, or claim information.

COMPLETION CRITERIA

Assets, sources, contributors, ownership, licenses, restrictions, approvals, notices, evidence, publication, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve provenance, licenses, contracts, approvals, attribution, claims, decisions, evidence, exceptions, and open gaps. Hand off to Legal, Licensing, Procurement, Records, Security, Privacy, Engineering, and Release owners.

RELATED CODEX SECTIONS

004 Evidence and Truth; 219 Legal; 221 Policy; 224 Licensing; 225 Records Management; 348 Procurement; 388 Version Management; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence requirements, and the integration roadmap. Exact jurisdictions, contributor agreements, source inventory, license policy, data and model rights, and acceptance authority remain unresolved.

224

SECTION 224 - LICENSING

CODEX/03_SECTIONS/224_LICENSING.txt

SHA-256 4ea7ea16632553d7967248b54cd46209ef5aa408f4ec8bc0e4a3f2a53a0b3d26

5,663 BYTES / PRESERVED SOURCE IDENTITY

SECTION 224 — Licensing

STATUS: CONTENT-COMPLETE

PURPOSE

Select, record, comply with, monitor, and retire licenses governing software, data, content, services, models, hardware, and other received or distributed material.

REQUIRED OUTCOME

A license inventory and compliance model with asset, version, licensor, rights, restrictions, notices, obligations, consumers, distribution, approval, renewal, exception, and evidence.

WHY THIS MATTERS

Licenses create conditions on use, modification, distribution, attribution, disclosure, data, support, and termination. Package presence or a license file does not prove compatibility, notice delivery, or compliance in the shipped context.

APPLICABILITY

Apply when the work consumes, modifies, embeds, distributes, trains on, publishes, or operates licensed software, data, content, models, APIs, hardware, or services.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no licensed or restricted asset, dependency, distribution, notice, renewal, or termination obligation is affected and an approved exact baseline supports it.

PREREQUISITES AND DEPENDENCIES

Use IP, Legal, Procurement, Security, Privacy, Supply Chain, Build, Release, Records, and Vendor baselines. Identify licensors, versions, transitive dependencies, recipients, jurisdictions, and approval authority.

REQUIRED INPUTS

Asset and dependency inventory, license text, version, source, licensor, rights, restrictions, notices, obligations, distribution model, consumers, contracts, renewals, exceptions, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners and counsel approve license selection, compatibility, distribution, attribution, disclosure, renewal, termination, exceptions, and risk. Humans resolve ambiguous or conflicting terms and approve shipped artifacts.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory dependencies, extract supplied license terms, compare obligations, draft notices and questions, and identify missing records. It must not interpret ambiguous terms conclusively, approve compatibility, remove notices, accept risk, or distribute material without qualified human authority.

APPLICABLE EXISTING AGENT ROLES

License Analyst, Legal Advisor, IP Analyst, Procurement Analyst, Researcher, Governance Reviewer, Verifier, Documentation Writer, Critic, and Memory Curator.

REQUIRED ACTIVITIES

Inventory direct and transitive assets; capture source and version; identify terms and obligations; assess compatibility and distribution; generate notices; approve exceptions; monitor change and renewal; validate shipped artifacts; handle termination or claim; preserve records.

HUMAN-AI INTERACTION REQUIREMENTS

Show asset, version, source, license, obligation, restriction, distribution, jurisdiction, evidence, and uncertainty. Require human or counsel review for ambiguous, reciprocal, data, model, export, termination, or high-impact terms.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Connect source, dependency, build, artifact, license, notice, consumer, distribution, and retirement records. Gate builds and releases on approved compatibility and retain notices and obligations with the exact artifact.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show license, asset, version, owner, obligations, distribution, notice, renewal, exception, evidence freshness, and authorized approve, restrict, replace, remediate, withdraw, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

License register; dependency and provenance map; compatibility assessment; notice and attribution package; approval and exception record; renewal and termination plan; test, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, version, source, licensor, license, contract, artifact, distribution, method, reviewer, result, and limitation. A package manager record or license filename proves only its captured metadata.

VERIFICATION AND ACCEPTANCE TESTS

Verify transitive coverage, exact version, license text, compatibility, notices, artifact inclusion, distribution, renewal, termination, and withdrawal. Test dependency update, missing metadata, license conflict, and build failure.

FAILURE STATES AND ANTI-PATTERNS

License assumed from package name, transitive dependency ignored, notices dropped, incompatible terms combined, expired entitlement, and distribution before approval.

RECOVERY AND RECONSIDERATION

Restrict affected artifact or distribution, preserve records, notify owners and counsel, replace or remediate material, correct notices, and assess downstream recipients. Reopen on dependency, license, contract, artifact, jurisdiction, or distribution change.

COMPLETION CRITERIA

Assets, versions, licensors, terms, obligations, artifacts, owners, approvals, notices, evidence, exceptions, recovery, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve license and dependency records, notices, approvals, exceptions, renewals, claims, artifacts, evidence, and open gaps. Hand off to IP, Legal, Procurement, Supply Chain, Build, Release, Records, and Governance.

RELATED CODEX SECTIONS

219 Legal; 220 Regulatory Compliance; 223 Intellectual Property; 225 Records Management; 348 Procurement; 388 Version Management; 397-400 Decommissioning and Disposal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, IP and legal context, and the integration roadmap. Exact asset inventory, license policy, contract terms, jurisdictions, notices, renewals, and acceptance authority remain unresolved.

225

SECTION 225 - RECORDS MANAGEMENT

CODEX/03_SECTIONS/225_RECORDS_MANAGEMENT.txt

SHA-256 d1071dcb82f9b8a0eeb1d7e1ce051ee0d30c6baca68f57e1e6da587cec1776fc

5,746 BYTES / PRESERVED SOURCE IDENTITY

SECTION 225 — Records Management

STATUS: CONTENT-COMPLETE

PURPOSE

Identify, classify, create, capture, protect, retain, retrieve, preserve, transfer, and dispose of records that provide evidence of obligations, decisions, rights, operations, or history.

REQUIRED OUTCOME

A records model with record classes, owners, authoritative copies, metadata, retention, legal hold, access, integrity, disposition, audit, continuity, and exception rules.

WHY THIS MATTERS

Records must remain usable and trustworthy for the period required, while unnecessary retention increases privacy, cost, and discovery risk. A backup, document, log, or database row is not automatically a record or an authoritative copy.

APPLICABILITY

Apply when information documents business, legal, regulatory, financial, safety, security, product, user, operational, or governance activity or must be preserved for accountability.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no record, hold, retention, evidence, right, audit, or disposition obligation is affected and an approved exact baseline supports the conclusion.

PREREQUISITES AND DEPENDENCIES

Use Legal, Regulatory, Privacy, Security, Data, Evidence, Audit, Backup, Archive, and Deletion baselines. Identify jurisdictions, owners, record classes, authoritative systems, holds, retention authority, and disposal approval.

REQUIRED INPUTS

Record classes, sources, metadata, owners, business and legal value, retention, access, integrity, holds, jurisdictions, disposition, transfers, systems, evidence, and exceptions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Records owners and counsel approve classification, retention, hold, access, transfer, disposition, and exceptions. Humans resolve whether a record is authoritative, whether a hold applies, and whether destruction is lawful and safe.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify authorized metadata, identify duplicates, draft inventories, summarize records, and propose retention or disposition queues. It must preserve provenance and not delete, alter, reclassify, disclose, or declare a record non-authoritative without explicit human authority and hold checks.

APPLICABLE EXISTING AGENT ROLES

Records Manager, Legal Advisor, Privacy Reviewer, Archivist, Researcher, Analyst, Governance Reviewer, Security Auditor, Verifier, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define record classes and authority; inventory sources; capture metadata; establish retention and access; apply holds; preserve integrity; enable retrieval; transfer or archive; review and dispose; audit decisions; handle exceptions and continuity.

HUMAN-AI INTERACTION REQUIREMENTS

Show record class, source, authority, owner, hold, retention, sensitivity, action, evidence, and uncertainty. Require human approval for hold, disclosure, reclassification, disposition, and destruction. Keep protected content out of unnecessary AI context.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate operational data, records, evidence, backups, archives, holds, indexes, access, and disposition. Preserve metadata and integrity across migration, provide legal hold override, and prevent retention automation from destroying held records.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show record class, owner, authoritative source, retention, hold, access, integrity, evidence freshness, disposition state, and authorized classify, preserve, search, restrict, transfer, and destroy controls.

REQUIRED OUTPUTS AND ARTIFACTS

Records schedule; classification and authority register; metadata and access model; hold and disposition procedures; transfer and archive plan; retrieval, audit, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify record, source, class, authority, version, metadata, integrity, owner, hold, retention, access, method, time, result, and limitation. A backup or export proves only its capture and declared completeness.

VERIFICATION AND ACCEPTANCE TESTS

Verify capture, metadata, search, access, integrity, hold, retention, transfer, archive, disposition, deletion, recovery, and audit. Test duplicate sources, migration, clock, hold arrival, restricted access, and failed disposition.

FAILURE STATES AND ANTI-PATTERNS

Records without owner or metadata, uncontrolled copies, hold bypass, premature deletion, indefinite retention, backup confused with record, and AI summary treated as the authoritative record.

RECOVERY AND RECONSIDERATION

Stop disposition, preserve affected copies, apply or extend holds, restore metadata, notify owners, and investigate loss or disclosure. Reassess legal, privacy, evidentiary, and continuity impact before resuming.

COMPLETION CRITERIA

Classes, sources, authority, owners, metadata, access, holds, retention, disposition, evidence, tests, exceptions, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve schedules, records, holds, disposition decisions, transfers, evidence, incidents, exceptions, and open gaps. Hand off to Legal, Compliance, Privacy, Audit, Data, Archive, Backup, Deletion, and Operations.

RELATED CODEX SECTIONS

004 Evidence and Truth; 219 Legal; 220 Regulatory Compliance; 223-224 IP and Licensing; 226 Audit; 391-395 Data Retention, Archive, Preservation, Disposal, and Deletion; 402-406 Review and Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence and lifecycle requirements, and the integration roadmap. Exact schedules, jurisdictions, record authority, holds, retention, archive, disposal, and acceptance evidence remain unresolved.

226

SECTION 226 - AUDIT

CODEX/03_SECTIONS/226_AUDIT.txt

SHA-256 a3c18d6701fc3bd1dd71844f84ec40e46e44ad0f5008d7a539579133d1391b3f

5,567 BYTES / PRESERVED SOURCE IDENTITY

SECTION 226 — Audit

STATUS: CONTENT-COMPLETE

PURPOSE

Provide independent, evidence-based examination of governance, risk, controls, operations, records, and claims, with findings that owners can act on.

REQUIRED OUTCOME

An audit scope, criteria, independence statement, evidence plan, sampling method, procedures, findings, management response, remediation, follow-up, and residual-risk record.

WHY THIS MATTERS

Audit can expose gaps between policy, design, operation, and outcome, but it loses value when scope is vague, evidence is curated, independence is compromised, or a clean report is treated as permanent assurance.

APPLICABILITY

Apply when independent assurance is needed for material controls, obligations, risks, financial or operational claims, safety, security, privacy, quality, or governance.

NON-APPLICABILITY CONDITIONS

Record non-applicability when the consequence and scope do not warrant independent audit and a documented review or test provides sufficient bounded evidence. Name the alternative, owner, date, and reopening condition.

PREREQUISITES AND DEPENDENCIES

Define objective, criteria, scope, period, independence, access, confidentiality, sampling, evidence retention, reporting, remediation authority, and follow-up owner.

REQUIRED INPUTS

Criteria, policies, controls, risks, systems, records, population, period, prior findings, incidents, owners, evidence, sampling frame, access, and acceptance expectations.

HUMAN RESPONSIBILITIES AND AUTHORITY

Audit leadership protects independence and approves scope, methods, findings, report, and follow-up. Owners provide truthful access and responses; management decides remediation and accepts residual risk without suppressing findings.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize evidence, compare criteria, sample authorized records, identify anomalies, draft workpapers, and summarize responses. It must preserve source, sample, period, method, independence, and uncertainty. It must not fabricate evidence, alter samples, issue an opinion, suppress a finding, or sign an attestation.

APPLICABLE EXISTING AGENT ROLES

Auditor, Researcher, Analyst, Verifier, Governance Reviewer, Security Auditor, Privacy Reviewer, Critic, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Set criteria and scope; confirm independence; plan sampling; obtain evidence; perform procedures; document exceptions; validate with owners; rate findings; report; agree remediation; follow up; retain workpapers; and reassess when scope changes.

HUMAN-AI INTERACTION REQUIREMENTS

Show criteria, population, sample, source, period, procedure, result, limitation, reviewer, and independence. Require human audit judgment for findings, opinion, severity, management response, and closure. Protect confidential evidence.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link criteria, controls, systems, records, evidence, samples, findings, responses, and remediation. Preserve read-only or immutable workpapers, segregation of duties, access audit, and reproducible sampling.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show audit scope, criteria, period, independence, progress, evidence coverage, findings, owner response, due dates, residual risk, and authorized assign, challenge, accept, remediate, escalate, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Audit charter; criteria and scope; independence statement; evidence and sampling plan; workpapers; findings and report; responses; remediation and follow-up; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify criterion, population, sample, source, period, procedure, auditor, result, exception, confidence, and limitation. A policy, screenshot, or management statement is bounded evidence and may require corroboration.

VERIFICATION AND ACCEPTANCE TESTS

Verify scope, criteria, independence, sample reproducibility, evidence integrity, finding traceability, management response, remediation, follow-up, and report approval. Test contradictory or missing evidence and late changes.

FAILURE STATES AND ANTI-PATTERNS

Audit by checklist, management-selected evidence only, scope drift, compromised independence, unsupported clean opinion, findings without owners, and closure without remediation evidence.

RECOVERY AND RECONSIDERATION

Preserve workpapers, disclose scope or evidence limitations, reopen affected findings, obtain independent review, and correct the report through controlled amendment. Reassess when systems, criteria, period, risks, or evidence change.

COMPLETION CRITERIA

Scope, criteria, independence, evidence, procedures, findings, responses, remediation, follow-up, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent audit approval and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve charter, workpapers, evidence references, findings, responses, reports, follow-up, exceptions, and open gaps. Hand off to Controls, Compliance, Legal, Records, Security, Privacy, Quality, and Governance.

RELATED CODEX SECTIONS

004 Evidence and Truth; 018 Risk Management; 220 Regulatory Compliance; 225 Records Management; 227 Internal Controls; 228-230 Quality; 402-404 Review and Postmortem.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence and control requirements, and the integration roadmap. Exact audit criteria, independence, sampling, reporting, jurisdiction, confidentiality, and acceptance authority remain unresolved.

227

SECTION 227 - INTERNAL CONTROLS

CODEX/03_SECTIONS/227_INTERNAL_CONTROLS.txt

SHA-256 caaa5d5091b8023abc0e940395141491c4a46e8a51e6f41305237363065d3d66

5,766 BYTES / PRESERVED SOURCE IDENTITY

SECTION 227 — Internal Controls

STATUS: CONTENT-COMPLETE

PURPOSE

Design and operate controls that prevent, detect, correct, and evidence errors, fraud, unauthorized action, material misstatement, policy failure, and loss of accountability.

REQUIRED OUTCOME

A control framework with objectives, risks, control owners, preventive and detective activities, segregation of duties, evidence, frequency, exceptions, testing, remediation, and management review.

WHY THIS MATTERS

Controls can exist on paper while failing in operation, creating false assurance. Overly manual, duplicated, unowned, or untested controls increase cost and still miss errors or abuse.

APPLICABILITY

Apply when governance, financial, operational, security, privacy, safety, quality, compliance, or trust outcomes require repeatable control.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no material risk, obligation, protected outcome, decision, transaction, or evidence need requires a control and an approved exact baseline documents the conclusion.

PREREQUISITES AND DEPENDENCIES

Use risk, requirements, policy, process, architecture, audit, compliance, security, privacy, quality, records, and operations baselines. Identify control objectives, owners, operators, evidence, and independent testing authority.

REQUIRED INPUTS

Objectives, risks, obligations, processes, systems, transactions, roles, controls, evidence, frequency, thresholds, exceptions, prior findings, incidents, users, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Control owners design and operate controls; management approves risk, control objectives, exceptions, compensating controls, and remediation. Segregation, independent review, and escalation are assigned according to consequence. Humans attest only to evidence they can support.

AI RESPONSIBILITIES AND BOUNDARIES

AI may map risks to controls, inspect authorized evidence, identify gaps or duplicates, draft procedures and tests, and summarize exceptions. It must preserve control, population, period, owner, and evidence boundaries. It must not self-certify, fabricate evidence, approve exceptions, alter control operation, or close findings without human authority.

APPLICABLE EXISTING AGENT ROLES

Controls Analyst, Auditor, Risk Analyst, Governance Reviewer, Security Auditor, Privacy Reviewer, Researcher, Tester, Verifier, Critic, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define objectives and risks; design controls; assign owners and segregation; establish evidence and cadence; implement; train; monitor; test design and operation; manage exceptions; remediate; report; review changes; and retire obsolete controls.

HUMAN-AI INTERACTION REQUIREMENTS

Show risk, objective, control, owner, population, period, evidence, result, exception, and limitation. Require human review for control design, attestation, exception, compensating control, remediation, and closure. Preserve independent challenge.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link risk, obligation, process, system, control, evidence, finding, exception, and remediation. Prefer preventive and detective controls with least privilege, segregation, immutable evidence, change impact, and safe manual fallback.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show control objective, risk, owner, status, frequency, evidence freshness, test result, exception, due date, and authorized approve, operate, override, remediate, escalate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Control framework; risk and control matrix; owner and segregation map; procedures and evidence plan; test and exception records; management report; decision, remediation, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify risk, objective, control, owner, population, period, system, method, evidence, reviewer, result, exception, and limitation. A procedure, configuration, or sample proves only the bounded claim and period.

VERIFICATION AND ACCEPTANCE TESTS

Verify design, implementation, operation, evidence, frequency, segregation, override, exception, remediation, change impact, and independent testing. Sample normal and boundary cases and test control failure and recovery.

FAILURE STATES AND ANTI-PATTERNS

Controls without owners, evidence theater, self-review, permanent exceptions, manual workarounds hidden from the control, duplicate controls, and green status based on completion rather than outcome.

RECOVERY AND RECONSIDERATION

Preserve evidence, assess affected population and period, apply compensating control, notify owners, remediate, and reopen unsupported assertions. Reassess after process, system, risk, regulation, personnel, or supplier change.

COMPLETION CRITERIA

Objectives, risks, controls, owners, segregation, evidence, testing, exceptions, remediation, decisions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent verification and management acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve risk-control matrix, evidence, tests, findings, exceptions, attestations, decisions, remediation, and open gaps. Hand off to Audit, Compliance, Legal, Security, Privacy, Quality, Operations, Finance, Records, and Governance.

RELATED CODEX SECTIONS

018 Risk Management; 220 Regulatory Compliance; 221 Policy; 225 Records Management; 226 Audit; 228-230 Quality; 301-330 Operations; 402-406 Review and Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, risk and evidence requirements, and the integration roadmap. Exact control framework, materiality, owners, testing, attestation, exception, jurisdiction, and acceptance authority remain unresolved.

228

SECTION 228 - QUALITY MANAGEMENT

CODEX/03_SECTIONS/228_QUALITY_MANAGEMENT.txt

SHA-256 9048dbe3249f6184fda6f86a08fbcfe56a07e0ee0bbdcfdee8aa41e6a05b16e6

5,585 BYTES / PRESERVED SOURCE IDENTITY

SECTION 228 — Quality Management

STATUS: CONTENT-COMPLETE

PURPOSE

Govern quality objectives, responsibilities, methods, evidence, defects, risks, improvement, and acceptance across the lifecycle.

REQUIRED OUTCOME

A quality-management system with quality characteristics, owners, criteria, measures, reviews, nonconformance, corrective action, escalation, records, and continuous-improvement decisions.

WHY THIS MATTERS

Quality is broader than passing tests or meeting schedule. Without explicit characteristics and owners, teams optimize visible output while defects, user harm, maintainability, accessibility, reliability, and evidence quality accumulate.

APPLICABILITY

Apply when a product, service, process, system, artifact, or decision has quality expectations, acceptance risk, repeatable work, or consequences from defect or variation.

NON-APPLICABILITY CONDITIONS

Record non-applicability only for a bounded, reversible artifact with no material quality claim, consumer, safety, reliability, compliance, or lifecycle consequence and an approved exact baseline.

PREREQUISITES AND DEPENDENCIES

Use requirements, architecture, risk, governance, quality characteristics, testing, operations, user research, support, and records baselines. Identify owner, consumers, acceptance authority, and improvement cadence.

REQUIRED INPUTS

Objectives, stakeholders, quality characteristics, requirements, risks, processes, controls, defects, measures, test results, incidents, feedback, resources, thresholds, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Quality owners define criteria and measures; accountable humans approve tradeoffs, release or acceptance, exceptions, corrective action, and residual risk. Independent reviewers challenge claims and cannot be replaced by a green metric.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize quality evidence, identify trends, draft checklists and tests, and propose improvements. It must preserve source, population, period, definition, and uncertainty. It must not redefine quality, suppress defects, approve release, or attest to conformance without human authority.

APPLICABLE EXISTING AGENT ROLES

Quality Manager, Quality Engineer, Quality Assurance Reviewer, Researcher, Analyst, Architect, Tester, Verifier, Critic, Governance Reviewer, Documentation Writer, and Memory Curator.

REQUIRED ACTIVITIES

Define quality and acceptance; map objectives and risks; assign owners; establish methods and measures; plan reviews and tests; record defects and nonconformance; correct and prevent recurrence; report trends; review changes; and improve the system.

HUMAN-AI INTERACTION REQUIREMENTS

Show quality definition, source, population, period, measure, threshold, result, limitation, owner, and decision. Require human review for material tradeoffs, acceptance, exceptions, and corrective-action closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link requirements, quality characteristics, controls, tests, defects, incidents, measures, decisions, and evidence. Preserve versioned baselines, independent review, traceability, and feedback loops that do not reward hiding defects.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show quality objectives, measures, findings, owners, evidence freshness, trends, risk, exceptions, corrective actions, and authorized accept, reject, escalate, remediate, reopen, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Quality charter; quality characteristics and criteria; measure catalog; review and test plan; defect and nonconformance register; corrective-action records; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify characteristic, definition, source, population, period, method, result, owner, reviewer, uncertainty, and limitation. A score, inspection, or passing test proves only its declared slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify quality definitions, measurement, sampling, traceability, defects, corrective action, independent review, acceptance, change impact, and feedback. Test missing evidence, metric drift, and conflicting signals.

FAILURE STATES AND ANTI-PATTERNS

Quality theater, vanity metrics, defect suppression, acceptance by schedule, unowned exceptions, measuring activity instead of outcome, and treating compliance paperwork as quality.

RECOVERY AND RECONSIDERATION

Mark unsupported claims, contain affected output, preserve evidence, correct defects, notify owners, and reassess acceptance. Reopen on changed requirements, users, risk, evidence, process, supplier, or lifecycle.

COMPLETION CRITERIA

Objectives, characteristics, owners, criteria, measures, evidence, reviews, defects, corrective action, exceptions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent assurance and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve quality baseline, measures, findings, decisions, evidence, corrective action, incidents, exceptions, and open gaps. Hand off to Quality Engineering, QA, Testing, Operations, Governance, Support, and Retirement.

CODEX/03_SECTIONS/228_QUALITY_MANAGEMENT.txt

RELATED CODEX SECTIONS

019 Requirements Engineering; 228-230 Quality Engineering and Assurance; 231-254 Testing; 301-330 Operations; 402-406 Review and Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, requirements and evidence context, and the integration roadmap. Exact quality model, measures, thresholds, materiality, authority, sector obligations, and acceptance evidence remain unresolved.

229

SECTION 229 - QUALITY ENGINEERING

CODEX/03_SECTIONS/229_QUALITY_ENGINEERING.txt

SHA-256 24cd9779c4f19b472365c92671de6fc701e60aaf9d5be418e666bdbd3add5c7c

5,544 BYTES / PRESERVED SOURCE IDENTITY

SECTION 229 — Quality Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Build quality into requirements, architecture, design, implementation, operation, and retirement through measurable characteristics, prevention, feedback, automation, and learning.

REQUIRED OUTCOME

A quality-engineering approach that links risks and quality characteristics to design choices, controls, tests, observability, defect prevention, maintainability, and measurable outcomes.

WHY THIS MATTERS

Finding defects late is expensive and can leave structural causes untouched. Quality engineering prevents teams from treating testing as the only quality activity or from automating checks that measure the wrong behavior.

APPLICABILITY

Apply when quality must be designed into software, hardware, data, processes, services, AI, or operations rather than inspected only after production.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when the artifact is low consequence, isolated, reversible, and has no material quality characteristic or consumer expectation beyond a documented baseline.

PREREQUISITES AND DEPENDENCIES

Use requirements, architecture, risk, quality management, testing, observability, operations, user research, and support evidence. Identify quality characteristics, constraints, environments, and feedback owners.

REQUIRED INPUTS

Quality goals, risks, requirements, architecture, user journeys, interfaces, data, dependencies, defect history, testability, observability, thresholds, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners set protected outcomes and tradeoffs; engineers design controls and testability; humans decide acceptance, exception, and residual risk. Specialists review safety, security, accessibility, performance, and domain quality where relevant.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized requirements and defect evidence, identify quality risks, draft tests and instrumentation, and propose design improvements. It must not substitute generated tests for user or domain judgment, redefine acceptance, hide uncertainty, or approve release.

APPLICABLE EXISTING AGENT ROLES

Quality Engineer, Architect, Researcher, Analyst, Implementer, Tester, Verifier, User Experience Reviewer, Security Auditor, Critic, Optimizer, and Documentation Writer.

REQUIRED ACTIVITIES

Define characteristics and risks; design for testability, observability, resilience, accessibility, security, and maintainability; automate repeatable checks; review defects and causes; measure outcomes; experiment safely; and feed lessons into requirements and architecture.

HUMAN-AI INTERACTION REQUIREMENTS

Show quality goal, risk, evidence, test or control, owner, assumption, result, and limitation. Require human review for generated tests, design tradeoffs, high-impact changes, and acceptance. Preserve exploratory and qualitative evidence.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Expose contracts, invariants, state, errors, telemetry, versioning, fault injection, rollback, and test seams. Keep production safety and privacy boundaries intact while enabling representative validation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show quality risk, characteristic, control, test coverage, defect trend, evidence freshness, owner, decision, and authorized accept, defer, remediate, experiment, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Quality strategy; characteristic and risk map; design and testability decisions; automation and observability plan; defect and cause analysis; experiment, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify quality characteristic, scenario, configuration, population, method, result, owner, environment, uncertainty, and limitation. Automation coverage is not behavioral coverage unless its scope is shown.

VERIFICATION AND ACCEPTANCE TESTS

Verify quality risks are linked to controls and tests; validate representative data and environments; exercise failure, accessibility, security, performance, recovery, and change paths; and independently reproduce material claims.

FAILURE STATES AND ANTI-PATTERNS

Test-only quality, brittle checks, coverage theater, unrepresentative fixtures, no user evidence, ignored defect causes, and optimization that harms protected characteristics.

RECOVERY AND RECONSIDERATION

Stop affected release, preserve evidence, diagnose cause, use rollback or containment, update design and tests, and reassess acceptance. Reopen on requirement, architecture, usage, dependency, or defect-pattern change.

COMPLETION CRITERIA

Quality characteristics, risks, design controls, tests, observability, owners, evidence, defects, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve quality strategy, design decisions, tests, instrumentation, defect history, evidence, experiments, and open gaps. Hand off to Quality Management, QA, Testing, Architecture, Operations, Support, and Governance.

RELATED CODEX SECTIONS

CODEX/03_SECTIONS/229_QUALITY_ENGINEERING.txt

019 Requirements Engineering; 020 Systems Engineering; 228 Quality Management; 230 Quality Assurance; 231-254 Testing; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, requirements and quality context, and the integration roadmap. Exact quality characteristics, domain risks, testability, measures, thresholds, and acceptance authority remain unresolved.

230

SECTION 230 - QUALITY ASSURANCE

CODEX/03_SECTIONS/230_QUALITY_ASSURANCE.txt

SHA-256 614de3a03e7bbb35ed76469e47480f84e8c8c6151ad1fd5903a402e2192835b9

5,185 BYTES / PRESERVED SOURCE IDENTITY

SECTION 230 — Quality Assurance

STATUS: CONTENT-COMPLETE

PURPOSE

Provide independent confidence that quality processes, controls, evidence, and outcomes satisfy their declared criteria within a bounded scope.

REQUIRED OUTCOME

A quality-assurance plan with independence, criteria, reviews, test oversight, evidence, nonconformance, escalation, release or acceptance input, and follow-up.

WHY THIS MATTERS

Assurance is weakened when the people producing an artifact are the only people judging it, when evidence is selected to confirm a conclusion, or when a passing review is treated as permanent confidence.

APPLICABILITY

Apply when independent review or assurance is needed for quality, safety, security, privacy, compliance, user impact, release, or material operational claims.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when the consequence is low, the scope is reversible, and a documented peer or automated check provides sufficient bounded confidence.

PREREQUISITES AND DEPENDENCIES

Define criteria, scope, independence, evidence access, reviewers, confidentiality, sampling, decision authority, and escalation. Use quality management, engineering, testing, audit, and governance baselines.

REQUIRED INPUTS

Requirements, quality criteria, risks, designs, processes, controls, test results, defects, incidents, user evidence, owners, environments, and acceptance expectations.

HUMAN RESPONSIBILITIES AND AUTHORITY

QA owners protect independence, plan reviews, challenge evidence, report findings, and recommend acceptance. Accountable owners decide release, exception, remediation, or residual risk; reviewers do not silently accept on their behalf.

AI RESPONSIBILITIES AND BOUNDARIES

AI may compare evidence to criteria, identify omissions, draft review questions, and summarize bounded findings. It must preserve reviewer independence, source, scope, and uncertainty. It must not issue assurance, suppress findings, fabricate review, or sign approval.

APPLICABLE EXISTING AGENT ROLES

Quality Assurance Reviewer, Quality Manager, Tester, Verifier, Auditor, Researcher, Analyst, Critic, Governance Reviewer, User Experience Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Set criteria and scope; confirm independence; review plans and artifacts; sample evidence; observe tests and operations; record nonconformance; communicate findings; verify corrective action; provide bounded assurance; and reopen on change.

HUMAN-AI INTERACTION REQUIREMENTS

Show criterion, evidence, method, reviewer, scope, result, limitation, and decision authority. Require human judgment for findings, assurance, release, exception, and closure. Keep confidential evidence controlled.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link requirements, controls, tests, observations, findings, decisions, and evidence. Preserve immutable review records, segregation, repeatable sampling, and traceable baselines.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show assurance scope, criteria, reviewer, evidence coverage, findings, exceptions, due dates, and authorized challenge, accept, remediate, escalate, reopen, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

QA charter; criteria and review plan; evidence and sampling record; findings and nonconformance; corrective-action follow-up; assurance statement; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify criterion, artifact, environment, sample, method, reviewer, time, result, limitation, and confidence. A review checklist or automated pipeline proves only its actual scope and operation.

VERIFICATION AND ACCEPTANCE TESTS

Verify independence, criteria, scope, evidence, sampling, test observation, finding traceability, corrective action, release input, and closure. Test contradictory evidence, stale baselines, and reviewer conflict.

FAILURE STATES AND ANTI-PATTERNS

Assurance by checkbox, self-approval, undisclosed conflict, selective evidence, green pipeline as release proof, and findings closed by assertion.

RECOVERY AND RECONSIDERATION

Withdraw unsupported assurance, preserve workpapers, obtain independent review, correct findings, and reassess release or acceptance. Reopen on changed evidence, scope, requirements, risk, or operation.

COMPLETION CRITERIA

Criteria, scope, independence, evidence, findings, decisions, limitations, follow-up, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent approval and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve reviews, evidence references, findings, decisions, exceptions, corrective actions, and open gaps. Hand off to Quality Management, Testing, Audit, Governance, Release, Operations, and Support.

RELATED CODEX SECTIONS

019 Requirements Engineering; 228 Quality Management; 229 Quality Engineering; 231-254 Testing; 226 Audit; 301-330 Operations; 402-404 Review.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, quality and evidence context, and the integration roadmap. Exact independence, criteria, sampling, assurance language, materiality, and acceptance authority remain unresolved.

231

SECTION 231 - SOFTWARE TESTING

CODEX/03_SECTIONS/231_SOFTWARE_TESTING.txt

SHA-256 061f7ad3b6ce050a73e542cb2c27ae7ad35d7b0c7316b5b28f7965718fd60091

5,295 BYTES / PRESERVED SOURCE IDENTITY

SECTION 231 — Software Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Evaluate software behavior against requirements, risks, contracts, quality characteristics, user needs, security, performance, operability, and recovery expectations.

REQUIRED OUTCOME

A software test strategy with scope, risks, levels, environments, data, oracles, automation, manual judgment, evidence, defects, coverage, exit criteria, and limitations.

WHY THIS MATTERS

Passing examples do not prove correct behavior across states, inputs, integrations, users, versions, or failures. Tests can also encode the wrong requirement, become stale, or pass while accessibility, security, performance, and recovery fail.

APPLICABILITY

Apply when software behavior, change, dependency, configuration, interface, model, or operational path can affect a declared outcome.

NON-APPLICABILITY CONDITIONS

Record non-applicability only for a bounded, low-consequence, non-executable artifact or a no-behavior change covered by an approved exact baseline.

PREREQUISITES AND DEPENDENCIES

Use requirements, architecture, risk, quality, user, security, data, environment, release, operations, and recovery baselines. Define test authority, data handling, environment, and acceptance owner.

REQUIRED INPUTS

Requirements, risks, interfaces, code, configuration, dependencies, data, users, platforms, environments, test oracles, prior defects, observability, recovery, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Test owners define scope, risk, evidence, and exit criteria; domain owners judge meaning; humans approve test data, production testing, exceptions, and release. Testers report failures without changing them to achieve a target.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft tests, fixtures, assertions, coverage analysis, and defect summaries from authorized context. It must not claim execution, fabricate results, use sensitive data without authority, or infer acceptance from generated tests or a green pipeline.

APPLICABLE EXISTING AGENT ROLES

Tester, Test Architect, Researcher, Analyst, Implementer, Verifier, Quality Assurance Reviewer, Security Auditor, User Experience Reviewer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define risks and levels; derive cases and oracles; prepare representative data and environments; execute and observe; record failures; analyze coverage and gaps; retest fixes; assess regression; report limitations; and preserve continuity.

HUMAN-AI INTERACTION REQUIREMENTS

Show requirement, case, data, environment, version, execution, result, evidence, and limitation. Require human review for test oracle, exploratory interpretation, high-impact failure, data use, and exit decision.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support deterministic fixtures, isolated environments, observability, fault injection, versioned tests, safe cleanup, reproducible execution, and separation of test and production authority.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show test scope, version, environment, coverage, failures, flaky state, evidence freshness, owner, exit criteria, and authorized run, pause, rerun, quarantine, accept, reject, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Test strategy and matrix; cases and fixtures; environment record; execution results; defect and coverage records; retest and exit decision; evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify requirement, case, version, data, environment, tool, execution, actor, time, result, evidence, coverage, and limitation. Test count or code coverage proves neither correctness nor absence of defect.

VERIFICATION AND ACCEPTANCE TESTS

Verify traceability, representative data, oracle, environment, execution integrity, failure recording, retest, flaky analysis, coverage, accessibility, security, performance, and recovery paths.

FAILURE STATES AND ANTI-PATTERNS

Testing only the happy path, assertion tautology, unrepresentative fixtures, flaky tests hidden, coverage theater, test data leakage, and release based on count rather than risk.

RECOVERY AND RECONSIDERATION

Preserve failing evidence, quarantine unreliable tests, reproduce in a controlled environment, correct code or test, and reassess exit criteria. Reopen on requirement, code, dependency, environment, data, or risk change.

COMPLETION CRITERIA

Scope, risks, cases, oracles, data, environments, evidence, defects, coverage, exit criteria, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending QA and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve test and environment versions, fixtures, results, defects, decisions, exceptions, and open gaps. Hand off to Unit, Integration, System, E2E, Acceptance, Performance, Release, Quality, and Operations owners.

RELATED CODEX SECTIONS

019 Requirements; 020 Systems Engineering; 228-230 Quality; 232-254 Test Levels and Specialized Testing; 301-330 Operations; 402-404 Review.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, requirements, quality, and evidence context, and the integration roadmap. Exact risk model, test environments, data, standards, exit authority, and acceptance evidence remain unresolved.

232

SECTION 232 - HARDWARE TESTING

CODEX/03_SECTIONS/232_HARDWARE_TESTING.txt

SHA-256 c370adfe98e5a0d8b36995baef4073b710cacb41258767028addaaadf60d9ac6

5,480 BYTES / PRESERVED SOURCE IDENTITY

SECTION 232 — Hardware Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Evaluate hardware behavior, interfaces, tolerances, safety, reliability, manufacturing variation, environmental response, and integration against declared requirements.

REQUIRED OUTCOME

A hardware-test strategy with representative units, fixtures, instrumentation, conditions, limits, safety controls, data integrity, failures, coverage, acceptance, and lifecycle limitations.

WHY THIS MATTERS

Hardware behavior varies with component lot, temperature, power, wear, assembly, environment, and interaction. A single bench result cannot prove field reliability, safety, or behavior outside its configuration.

APPLICABILITY

Apply when physical components, boards, devices, sensors, actuators, enclosures, interfaces, or manufacturing variation affect a requirement or operational outcome.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no physical behavior, interface, tolerance, safety, reliability, or manufacturing claim changes and an approved exact baseline applies.

PREREQUISITES AND DEPENDENCIES

Use hardware, embedded, safety, quality, manufacturing, environmental, security, and requirements baselines. Define units, fixtures, calibration, hazards, test authority, and acceptance owner.

REQUIRED INPUTS

Requirements, drawings, revisions, bill of materials, tolerances, units, environments, loads, fixtures, instrumentation, hazards, prior failures, data plan, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Hardware and safety owners approve conditions, hazards, destructive testing, sample size, deviations, and acceptance. Qualified testers protect people and equipment; owners decide whether a result represents the population.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize authorized specifications, test data, failure patterns, and calibration records; draft procedures and analysis; and propose bounded experiments. It must not control hazardous equipment, infer safety, alter data, or accept a unit without qualified human supervision.

APPLICABLE EXISTING AGENT ROLES

Hardware Tester, Hardware Engineer, Safety Reviewer, Quality Engineer, Researcher, Analyst, Observer, Implementer, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define requirements and risks; select representative units; verify fixtures and calibration; establish safe conditions; execute functional, boundary, environmental, durability, interface, and recovery tests; record failures; analyze variation; retest; and report limits.

HUMAN-AI INTERACTION REQUIREMENTS

Show unit, revision, fixture, calibration, condition, load, instrument, method, result, hazard, and limitation. Require human approval for hazardous, destructive, field, or safety-relevant testing and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide test points, instrumentation, safe isolation, fixture control, unit identity, environmental control, traceable data, and recovery from power, sensor, actuator, and communication failure.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show unit population, revision, condition, active test, hazard, calibration, failure, evidence freshness, owner, and authorized pause, stop, quarantine, accept, retest, and dispose controls.

REQUIRED OUTPUTS AND ARTIFACTS

Hardware test plan; unit and fixture record; calibration and condition record; results and failures; variation and coverage analysis; retest, acceptance, safety, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify unit, revision, lot, fixture, calibration, condition, load, instrument, method, time, result, operator, and limitation. A bench pass or certification proves only its declared conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify fixtures, calibration, unit identity, safety, functional and boundary behavior, environmental limits, variation, repeatability, failure response, recovery, and data integrity. Reproduce material failures independently.

FAILURE STATES AND ANTI-PATTERNS

Uncalibrated instruments, nonrepresentative unit, unsafe setup, cherry-picked result, data alteration, tolerance ignored, destructive test without authority, and bench behavior treated as field proof.

RECOVERY AND RECONSIDERATION

Stop unsafe testing, isolate unit and fixture, preserve data, investigate, repair or replace under authority, and reassess population and acceptance. Reopen on revision, lot, environment, requirement, or failure-pattern change.

COMPLETION CRITERIA

Units, conditions, fixtures, requirements, hazards, results, failures, coverage, evidence, acceptance, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve unit and fixture identity, calibration, data, failures, decisions, evidence, exceptions, and open gaps. Hand off to Hardware, Embedded, Manufacturing, Safety, Quality, Release, Operations, and Disposal.

RELATED CODEX SECTIONS

058 Hardware Engineering; 060 Firmware Engineering; 199 Hardware Security; 228-230 Quality; 231 Software Testing; 233-242 Test Levels; 350-354 Manufacturing; 398-400 Disposal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, hardware, safety, quality, and evidence context, and the integration roadmap. Exact units, standards, environment, calibration, sample, hazards, and acceptance authority remain unresolved.

233

SECTION 233 - UNIT TESTING

CODEX/03_SECTIONS/233_UNIT_TESTING.txt

SHA-256 2a7714994b5c13231d5625ad258cda68aa769a8fe2aff508c9697d954f5eb7f1

4,732 BYTES / PRESERVED SOURCE IDENTITY

SECTION 233 — Unit Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Verify isolated functions, modules, classes, components, or hardware units against explicit behavior, boundaries, invariants, errors, and contracts.

REQUIRED OUTCOME

Fast, deterministic unit tests with clear subject, fixture, oracle, boundary cases, failure evidence, ownership, maintenance, and known limits.

WHY THIS MATTERS

Unit tests give precise feedback but can pass while integration, environment, user, security, or performance behavior fails. Brittle or tautological tests create false confidence and slow useful change.

APPLICABILITY

Apply when an isolated unit has meaningful behavior, contract, risk, or defect history that can be tested independently.

NON-APPLICABILITY CONDITIONS

Record non-applicability when the unit has no independent behavior or an approved higher-level test is more truthful and sufficient for the declared risk.

PREREQUISITES AND DEPENDENCIES

Use unit contract, requirements, code or hardware revision, dependencies, fixtures, mocks or stubs, test framework, and ownership. Avoid replacing meaningful integration with mocks.

REQUIRED INPUTS

Unit, contract, inputs, outputs, errors, invariants, boundaries, dependencies, fixture data, test environment, prior defects, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define behavior and risk; implementers maintain tests; reviewers challenge oracles and isolation; humans decide whether a unit-level result is enough for the larger claim.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft cases and fixtures, identify boundary gaps, and explain failures from authorized code and output. It must not claim execution, fabricate results, overuse mocks, or change the contract to make tests pass without human review.

APPLICABLE EXISTING AGENT ROLES

Tester, Implementer, Analyst, Verifier, Quality Engineer, Critic, Researcher, and Documentation Writer.

REQUIRED ACTIVITIES

Define oracle; cover normal, boundary, invalid, error, and state paths; isolate dependencies honestly; execute deterministically; diagnose failures; remove flakiness; retest; review coverage and limits; and preserve results.

HUMAN-AI INTERACTION REQUIREMENTS

Show unit, version, fixture, oracle, dependency substitution, execution, result, and limitation. Require human review for generated assertions, behavior interpretation, and test quarantine.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Expose stable contracts and deterministic seams without weakening encapsulation or production behavior. Keep fixtures versioned and prevent test-only paths from becoming security or operational bypasses.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show unit, suite, version, pass or fail, flakiness, coverage, owner, evidence freshness, and authorized run, quarantine, fix, rerun, and accept controls.

REQUIRED OUTPUTS AND ARTIFACTS

Unit-test matrix; fixtures; suite and environment record; results; failure and flakiness register; coverage and limitation report; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify unit, version, fixture, oracle, framework, environment, execution, result, time, actor, and limitation. Pass count or line coverage proves only its declared cases.

VERIFICATION AND ACCEPTANCE TESTS

Verify oracle quality, boundary and error coverage, deterministic execution, honest dependency isolation, failure reproduction, flakiness handling, and traceability to risk or contract.

FAILURE STATES AND ANTI-PATTERNS

Tautological assertions, over-mocking, private implementation fixation, flaky tests hidden, no invalid input, and unit pass treated as system correctness.

RECOVERY AND RECONSIDERATION

Preserve failure, isolate test or code cause, correct the contract or implementation, remove unsafe mock, and rerun. Reopen on unit, dependency, contract, or risk change.

COMPLETION CRITERIA

Unit, contract, fixtures, oracle, cases, results, failures, coverage, limits, owner, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve test and fixture versions, results, failures, decisions, exceptions, and open gaps. Hand off to Integration, System, Regression, Quality, Release, and Maintenance owners.

RELATED CODEX SECTIONS

019 Requirements; 229 Quality Engineering; 231 Software Testing; 234 Integration Testing; 238 Regression Testing; 386 Technical Debt Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, testing and quality context, and the integration roadmap. Exact framework, coverage, contract, fixture, flakiness, and acceptance rules remain unresolved.

234

SECTION 234 - INTEGRATION TESTING

CODEX/03_SECTIONS/234_INTEGRATION_TESTING.txt

SHA-256 c6a1ad27c9946ea2693d03bc81dd39c4679c2158a1af54e9985542efced86bae

4,997 BYTES / PRESERVED SOURCE IDENTITY

SECTION 234 — Integration Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Verify that connected components exchange data, control, identity, errors, timing, and recovery behavior according to their contracts.

REQUIRED OUTCOME

An integration-test model with participating versions, interfaces, fixtures, environments, data, dependency behavior, observability, failure paths, results, and limitations.

WHY THIS MATTERS

Components may pass isolated tests while disagreeing on schema, timing, identity, retries, errors, versions, or ownership. Integration evidence is bounded by the participants and conditions actually exercised.

APPLICABILITY

Apply when two or more components, services, devices, teams, providers, data stores, or processes interact through a contract.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no interaction, interface, dependency, data exchange, or control boundary changes and an approved exact baseline covers the work.

PREREQUISITES AND DEPENDENCIES

Use interface contracts, versions, data schemas, identity, environments, provider stubs or sandboxes, observability, recovery, and test-data controls.

REQUIRED INPUTS

Participants, interface, schema, versions, identities, data, timing, retries, failure behavior, environment, observability, prior defects, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Interface owners define contracts and compatibility; testers control environments and data; humans judge semantic behavior, provider risk, exceptions, and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft contract cases, fixtures, dependency matrices, and failure analysis from authorized artifacts. It must not claim external execution, alter provider or production state, expose data, or infer compatibility from syntax alone.

APPLICABLE EXISTING AGENT ROLES

Tester, Integration Engineer, Architect, Analyst, Implementer, Verifier, Quality Engineer, Security Auditor, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Map participants and contracts; choose representative versions and data; exercise normal, boundary, timeout, retry, duplicate, ordering, schema, identity, and failure paths; observe; diagnose; retest; and record compatibility limits.

HUMAN-AI INTERACTION REQUIREMENTS

Show participants, versions, contract, identity, data, environment, execution, result, and limitation. Require human approval for external or production dependency testing, data access, and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support contract versioning, test doubles with declared fidelity, isolated environments, correlation, deterministic cleanup, fault injection, retries, idempotency, timeout, and rollback.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show integration pair, versions, contract, status, failures, evidence freshness, owner, dependency health, and authorized run, pause, substitute, rollback, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Integration matrix; contract and fixture record; environment and dependency record; results and failure register; compatibility and coverage report; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify participants, versions, interface, data, environment, method, execution, result, error, time, actor, and limitation. A schema match or mock pass proves only the exercised contract slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify serialization, semantics, identity, authorization, ordering, retries, timeout, duplicate, version skew, provider loss, rollback, observability, and cleanup. Reproduce material failures independently.

FAILURE STATES AND ANTI-PATTERNS

Testing only happy interactions, mocks with false fidelity, version skew ignored, data semantics unverified, retry storms, and integration pass treated as end-to-end acceptance.

RECOVERY AND RECONSIDERATION

Preserve traces, isolate failing participant, use approved fallback, correct contract or component, and rerun. Reopen on interface, schema, version, provider, data, or operational change.

COMPLETION CRITERIA

Participants, contracts, versions, cases, data, results, failures, coverage, limits, owners, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve integration and dependency versions, fixtures, traces, results, decisions, exceptions, and open gaps. Hand off to System, E2E, API, Event, Release, Operations, and Quality owners.

RELATED CODEX SECTIONS

034 Integration Architecture; 035 API Architecture; 036 Event-Driven Architecture; 231 Software Testing; 233 Unit Testing; 235-238 System, E2E, Acceptance, and Regression Testing; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, integration architecture, and testing context. Exact contracts, providers, data, environment, compatibility, failure, and acceptance authority remain unresolved.

235

SECTION 235 - SYSTEM TESTING

CODEX/03_SECTIONS/235_SYSTEM_TESTING.txt

SHA-256 fc27e1fe440390bda5f7c39b701f5696e43b72105f8a8d3856a78e01e220db92

5,045 BYTES / PRESERVED SOURCE IDENTITY

SECTION 235 — System Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Evaluate the assembled system against functional, nonfunctional, operational, security, safety, user, data, and lifecycle requirements in a representative environment.

REQUIRED OUTCOME

A system-test baseline with versioned system, configuration, interfaces, data, users, scenarios, environment, results, defects, coverage, acceptance, and limitations.

WHY THIS MATTERS

System behavior emerges from interactions among components, configuration, people, providers, timing, and operations. Unit or integration success does not prove that the whole system meets its purpose or fails safely.

APPLICABILITY

Apply when an assembled system or service has meaningful functional, quality, safety, security, operational, or user outcomes.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no assembled behavior, configuration, user, interface, or lifecycle claim changes and a verified exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use requirements, architecture, operational concept, quality, risk, test levels, data, identity, environments, observability, recovery, and acceptance baselines.

REQUIRED INPUTS

System version, configuration, requirements, scenarios, users, data, dependencies, environment, quality risks, test oracles, prior findings, recovery, and exit criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

System owners define outcomes and acceptance; testers execute and report; domain, security, safety, accessibility, and operations owners judge their concerns. Humans approve production-like data, exceptions, and release.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft scenarios, generate bounded data, analyze results, and identify gaps from authorized context. It must not claim full-system coverage, use uncontrolled data, alter environments, or accept release without human judgment.

APPLICABLE EXISTING AGENT ROLES

System Tester, Systems Engineer, Architect, Quality Engineer, Security Auditor, User Experience Reviewer, Analyst, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Baseline system; select representative scenarios and users; exercise functional, boundary, failure, security, safety, performance, accessibility, recovery, and operational paths; record defects; retest; assess coverage and limits; obtain owner decisions.

HUMAN-AI INTERACTION REQUIREMENTS

Show system version, configuration, scenario, user, data, environment, execution, result, evidence, and limitation. Require human review for safety, security, user, recovery, and exit decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support representative environments, controlled dependencies, observability, fault injection, reset, data isolation, version traceability, rollback, and safe cleanup.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show system baseline, scenario coverage, failures, risks, evidence freshness, owner, exit state, and authorized run, pause, quarantine, remediate, accept, reject, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

System-test plan and matrix; version and configuration record; scenarios and data; results and defect register; coverage and limitation report; exit and acceptance; evidence and continuity records.

EVIDENCE REQUIREMENTS

Identify system, version, configuration, scenario, data, environment, method, execution, result, time, actor, coverage, and limitation. A green run proves only the executed scenarios and conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify functional and nonfunctional behavior, boundary and failure paths, user and operator journeys, security, safety, accessibility, recovery, observability, and reproducibility.

FAILURE STATES AND ANTI-PATTERNS

Environment drift, scenario theater, production data leakage, only happy-path coverage, system pass despite known operational gaps, and defects hidden as test exclusions.

RECOVERY AND RECONSIDERATION

Preserve state and failure evidence, reset or isolate environment, remediate, retest, and reassess exit. Reopen on system, configuration, dependency, user, requirement, or risk change.

COMPLETION CRITERIA

System, configuration, scenarios, data, results, failures, coverage, exit criteria, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending QA and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve system baseline, data, environment, results, defects, decisions, evidence, exceptions, and open gaps. Hand off to E2E, Acceptance, Regression, Performance, Security, Operations, and Release.

RELATED CODEX SECTIONS

020 Systems Engineering; 029-044 Architecture; 231 Software Testing; 234 Integration Testing; 236-242 Specialized Testing; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, systems and quality context, and the integration roadmap. Exact system, environment, users, data, scenario, risk, exit, and acceptance requirements remain unresolved.

236

SECTION 236 - END TO END TESTING

CODEX/03_SECTIONS/236_END_TO_END_TESTING.txt

SHA-256 9da0c68fca5c32a63fca5719f3fd18d77fcbfeb3a5286746902204a5decdc59c

4,972 BYTES / PRESERVED SOURCE IDENTITY

SECTION 236 — End-to-End Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Validate complete user, operator, data, control, and recovery journeys across system boundaries from initiating action through observable outcome.

REQUIRED OUTCOME

An end-to-end test model with journey, actors, versions, dependencies, data, environment, expected outcome, evidence, failure and recovery behavior, and user acceptance limits.

WHY THIS MATTERS

Users experience journeys, not components. A complete flow can fail through identity, data, timing, support, communication, deployment, or recovery even when each internal test passes.

APPLICABILITY

Apply when a material outcome crosses multiple components, teams, providers, roles, or lifecycle states and must be validated as experienced.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no cross-boundary journey or material user or operator outcome changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use user research, requirements, architecture, integration, system, identity, data, support, operations, and recovery baselines. Define representative actors, data, environment, and safe test authority.

REQUIRED INPUTS

Journey, actors, roles, versions, components, providers, data, states, expected outcomes, accessibility, failure, support, recovery, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners and users define meaningful outcomes; testers execute; domain, accessibility, security, and operations reviewers judge their concerns. Humans approve production-like data and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft journeys, fixtures, assertions, and analysis from authorized sources. It must not claim user experience, accessibility, or end-to-end success without actual representative execution and human review.

APPLICABLE EXISTING AGENT ROLES

User Experience Reviewer, Tester, Systems Engineer, Analyst, Researcher, Implementer, Verifier, Accessibility Reviewer, Security Auditor, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Select critical journeys; define actors and states; prepare data and dependencies; execute normal, boundary, error, support, accessibility, security, and recovery paths; observe user and system results; remediate; retest; and preserve limitations.

HUMAN-AI INTERACTION REQUIREMENTS

Show journey, actor, data, version, environment, step, result, evidence, and limitation. Require human review for user meaning, accessibility, safety, recovery, and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support trace correlation, deterministic reset, dependency substitution with fidelity labels, state cleanup, user-visible errors, support hooks, audit, rollback, and safe test data.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show journey, current step, actor, dependency, result, failures, evidence freshness, owner, and authorized run, pause, recover, accept, reject, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Journey catalog; end-to-end plan; actor and data record; environment and dependency record; results, defects, user findings, recovery, acceptance, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify journey, actor, version, step, dependency, data, environment, method, result, time, observer, and limitation. Automation completion proves only execution, not meaningful user outcome.

VERIFICATION AND ACCEPTANCE TESTS

Verify journey semantics, data, identity, authorization, communication, accessibility, errors, support, recovery, cleanup, and independent reproduction. Include representative and adverse conditions.

FAILURE STATES AND ANTI-PATTERNS

Synthetic journey unlike real use, missing support or recovery, hidden manual steps, dependency stubs mistaken for reality, and end-to-end pass used to hide poor component evidence.

RECOVERY AND RECONSIDERATION

Preserve trace and user impact, reset safely, remediate, provide support, and rerun. Reopen on journey, user, dependency, version, policy, or operational change.

COMPLETION CRITERIA

Journeys, actors, dependencies, data, outcomes, failures, recovery, user evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending user and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve journey and environment versions, traces, defects, user findings, decisions, evidence, exceptions, and open gaps. Hand off to Acceptance, Regression, Performance, Support, Release, and Operations.

RELATED CODEX SECTIONS

001 Research; 019 Requirements; 020 Systems Engineering; 231-235 Testing; 237 Acceptance Testing; 238 Regression Testing; 301-330 Operations.

CODEX/03_SECTIONS/236_END_TO_END_TESTING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, user, system, and testing context, and the integration roadmap. Exact journeys, users, environments, data, accessibility, support, recovery, and acceptance authority remain unresolved.

237

SECTION 237 - ACCEPTANCE TESTING

CODEX/03_SECTIONS/237_ACCEPTANCE_TESTING.txt

SHA-256 6d92eb5e997dd2f5fbff251b3e5cb806a8b65fe66c78f4154be9b1d9569dfea9

5,268 BYTES / PRESERVED SOURCE IDENTITY

SECTION 237 — Acceptance Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Enable authorized users, owners, operators, and affected stakeholders to determine whether a delivered capability meets its intended outcome and is acceptable for the declared scope.

REQUIRED OUTCOME

An acceptance model with decision owner, user or stakeholder scenarios, criteria, evidence, exceptions, unresolved issues, accessibility, operational readiness, and explicit accept, reject, or conditional decision.

WHY THIS MATTERS

Technical pass status cannot decide whether the capability solves the right problem, fits actual work, or is acceptable to those accountable for consequences. Acceptance without authority or honest limitations creates false launch confidence.

APPLICABILITY

Apply before a product, service, process, system, or change is accepted for use, release, transition, funding, or operational ownership.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when the change has no user, owner, operational, contractual, or material outcome effect and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use requirements, user research, system and E2E results, quality, accessibility, security, operations, support, training, recovery, and owner authority.

REQUIRED INPUTS

Objective, acceptance owner, users, scenarios, criteria, evidence, test results, known defects, risks, exceptions, training, support, deployment, and recovery state.

HUMAN RESPONSIBILITIES AND AUTHORITY

Authorized humans perform or witness acceptance, interpret user and domain meaning, approve exceptions, and decide release or rejection. Users can report mismatch or harm; AI cannot accept on their behalf.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize criteria, facilitate scenarios, summarize evidence, and surface unmet conditions. It must not simulate stakeholder consent, declare acceptance, minimize defects, or make a release decision without accountable human authority.

APPLICABLE EXISTING AGENT ROLES

User Experience Reviewer, Tester, Verifier, Quality Assurance Reviewer, Researcher, Analyst, Operations Reviewer, Accessibility Reviewer, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Confirm authority and criteria; select representative users and scenarios; execute; observe comprehension and outcome; record evidence and defects; assess operations and recovery; resolve or accept exceptions; document decision and next review.

HUMAN-AI INTERACTION REQUIREMENTS

Show criterion, actor, scenario, evidence, result, exception, impact, reviewer, and decision authority. Require accessible human participation and explicit approval or rejection. Preserve dissent and conditional acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support acceptance environments, realistic data, user feedback, traceability, defects, configuration identity, training, support, rollback, and post-acceptance monitoring.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show criteria, scenarios, participants, results, findings, risks, exceptions, readiness, evidence freshness, and authorized accept, reject, condition, defer, remediate, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Acceptance charter; criteria and scenario set; participant and authority record; evidence and findings; exception and conditional decision; readiness, support, recovery, decision, and continuity records.

EVIDENCE REQUIREMENTS

Identify criterion, actor, scenario, version, environment, method, result, observer, decision, limitation, and dissent. A signature or survey response proves only its bounded participation and claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify authority, representative use, accessibility, outcome, defect visibility, operational readiness, support, recovery, evidence, and decision traceability. Confirm accepted limitations and review date.

FAILURE STATES AND ANTI-PATTERNS

Proxy acceptance, rushed sign-off, silent dissent, acceptance by technical team alone, hidden exceptions, unrealistic users or data, and conditional approval without owner or expiry.

RECOVERY AND RECONSIDERATION

Pause transition, preserve decision and evidence, support users, remediate mismatch, and reconvene the authority. Reopen on material defect, user harm, changed scope, or failed operation.

COMPLETION CRITERIA

Authority, criteria, users, scenarios, evidence, defects, exceptions, decision, readiness, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending authorized acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve acceptance evidence, participant and decision records, exceptions, readiness, support, recovery, and open gaps. Hand off to Release, Operations, Support, Quality, Governance, and Product owners.

RELATED CODEX SECTIONS

001 User Research; 019 Requirements; 230 Quality Assurance; 231 Software Testing; 235-236 System and E2E Testing; 238 Regression Testing; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, user, quality, evidence, and operational context, and the integration roadmap. Exact users, authority, criteria, accessibility, readiness, support, and acceptance evidence remain unresolved.

238

SECTION 238 - REGRESSION TESTING

CODEX/03_SECTIONS/238_REGRESSION_TESTING.txt

SHA-256 5657e2055c644e3f9fade43fa0e363f72b19ead31fca27b8df1da5ff995b087a

4,874 BYTES / PRESERVED SOURCE IDENTITY

SECTION 238 — Regression Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Detect unintended change to previously working behavior, protected characteristics, interfaces, data, security, performance, operations, and user outcomes after modification.

REQUIRED OUTCOME

A risk-based regression suite and review process tied to change impact, versions, environments, defects, coverage, exclusions, results, and owner decision.

WHY THIS MATTERS

Change can break behavior outside the edited area through shared dependencies, data, configuration, timing, policy, or user journeys. A large suite can still miss the changed risk or hide flaky and obsolete cases.

APPLICABILITY

Apply when code, hardware, data, configuration, dependency, policy, model, provider, environment, or operational change may affect existing behavior.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when impact analysis shows no behavior, contract, risk, user, or operational change and an approved exact baseline supports it.

PREREQUISITES AND DEPENDENCIES

Use change impact, requirements, architecture, defect history, unit, integration, system, E2E, performance, security, and acceptance baselines.

REQUIRED INPUTS

Change, affected paths, dependencies, prior defects, critical journeys, suite, fixtures, environment, versions, exclusions, risk, and exit criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Change and test owners define impact and scope; humans approve exclusions, quarantine, release, and residual risk. Domain owners decide whether behavior remains meaningful.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze diffs and dependency graphs, propose regression cases, prioritize suites, and summarize failures. It must not declare “no impact,” remove tests, hide flakiness, or approve release without human review.

APPLICABLE EXISTING AGENT ROLES

Tester, Test Analyst, Architect, Implementer, Quality Engineer, Verifier, Researcher, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Classify change; map affected risks and journeys; select and update cases; run representative suites; analyze new, missing, and flaky failures; retest fixes; review exclusions; and record decision and limits.

HUMAN-AI INTERACTION REQUIREMENTS

Show change, impact evidence, selected cases, exclusions, versions, result, and limitation. Require human review for impact classification, test removal, exception, and release.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Make dependencies, contracts, versions, feature flags, migrations, configuration, and telemetry traceable. Support selective and full regression, deterministic reset, and safe rollback.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show change, impact, suite, coverage, failures, flaky state, exclusions, evidence freshness, owner, and authorized run, quarantine, expand, accept, reject, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Impact analysis; regression selection; suite and fixture versions; results and defect record; exclusion and exception record; retest, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify change, baseline, case, version, environment, fixture, execution, result, time, method, owner, coverage, and limitation. A green rerun proves only the selected suite and conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify impact analysis, critical path coverage, prior defect coverage, changed dependency behavior, exclusions, flakiness, performance, security, recovery, and independent reproduction.

FAILURE STATES AND ANTI-PATTERNS

Run everything without impact reasoning, run nothing for a “small” change, obsolete cases, flaky tests ignored, exclusions hidden, and regression pass treated as full acceptance.

RECOVERY AND RECONSIDERATION

Preserve failure, halt affected release, expand scope, correct code or test, and rerun. Reopen on further change, new defect, dependency drift, environment change, or user impact.

COMPLETION CRITERIA

Change, impact, cases, fixtures, results, failures, exclusions, coverage, owners, evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve baseline and suite versions, impact, results, defects, exceptions, decisions, evidence, and open gaps. Hand off to Release, Quality, Unit, Integration, System, E2E, Performance, and Operations.

RELATED CODEX SECTIONS

019 Requirements; 229 Quality Engineering; 231-237 Testing Levels; 239-242 Performance, Load, Stress, and Scalability; 388 Version Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, change and testing context, and the integration roadmap. Exact impact model, critical paths, environment, suite, exclusions, thresholds, and acceptance authority remain unresolved.

239

SECTION 239 - PERFORMANCE TESTING

CODEX/03_SECTIONS/239_PERFORMANCE_TESTING.txt

SHA-256 3ac21f6bb4ededе592f7571040666dd8f69e60619f402a88d0b9edb4573bb06a

5,076 BYTES / PRESERVED SOURCE IDENTITY

SECTION 239 — Performance Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Measure responsiveness, throughput, resource use, latency, capacity behavior, and user or operator experience against declared performance objectives and conditions.

REQUIRED OUTCOME

A performance-test model with workload, scenarios, service levels, environment, instrumentation, baselines, variability, bottleneck analysis, evidence, and limits.

WHY THIS MATTERS

Performance is workload and environment dependent. A fast sample or average can hide tail latency, contention, resource exhaustion, degraded user experience, or a provider bottleneck.

APPLICABILITY

Apply when response time, throughput, resource use, capacity, cost, or user experience is material to the capability.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when performance has no material user, operator, safety, cost, service, or scale consequence and an approved exact baseline covers the change.

PREREQUISITES AND DEPENDENCIES

Use performance objectives, architecture, workload, environment, telemetry, data, capacity, cost, reliability, and operations baselines. Define safe test authority and stop conditions.

REQUIRED INPUTS

Workload, scenarios, users, data, service levels, latency percentiles, throughput, resource limits, environment, dependencies, instrumentation, baseline, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define meaningful performance and tradeoffs; testers control load and safety; operators approve production-like testing; humans accept performance risk and cost.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft workloads, analyze traces, identify bottlenecks, and propose experiments. It must not generate unsafe load, claim generality from one environment, alter production, or optimize a metric while harming user outcomes without human review.

APPLICABLE EXISTING AGENT ROLES

Performance Engineer, Tester, Architect, Analyst, Observer, Optimizer, Verifier, Quality Engineer, Operations Engineer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define objectives and workload; baseline; prepare data and environment; execute normal and boundary tests; capture latency distribution, throughput, resource, cost, and user evidence; analyze bottlenecks; remediate; rerun; document limits.

HUMAN-AI INTERACTION REQUIREMENTS

Show workload, environment, version, dependency, metric definition, percentile, resource, cost, result, and limitation. Require human approval for load generation, production testing, resource changes, and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide observability, correlation, resource isolation, realistic data, capacity controls, backpressure, graceful degradation, caching semantics, and safe test cleanup.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show workload, environment, latency, throughput, resource, cost, bottleneck, evidence freshness, owner, and authorized start, pause, stop, tune, rollback, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Performance objectives; workload and scenario model; environment and instrumentation record; baseline and results; bottleneck and remediation record; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify version, workload, data, environment, dependency, metric, percentile, method, time, result, variability, and limitation. An average or one run proves only its observed slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify workload realism, instrument calibration, repeatability, latency distribution, throughput, resource, cost, user impact, degradation, recovery, and independent reproduction.

FAILURE STATES AND ANTI-PATTERNS

Benchmark theater, average-only reporting, synthetic workload unlike use, warm-cache bias, ignored tail latency, unsafe load, and optimization that shifts cost or harm elsewhere.

RECOVERY AND RECONSIDERATION

Stop unsafe test, preserve traces, recover resources, isolate bottleneck, roll back harmful tuning, and reassess targets. Reopen on workload, version, environment, dependency, user, or objective change.

COMPLETION CRITERIA

Objectives, workloads, environments, metrics, results, variability, bottlenecks, owners, evidence, limits, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve workload and environment versions, traces, baselines, results, tuning, decisions, evidence, exceptions, and open gaps. Hand off to Capacity, Load, Stress, Scalability, Reliability, Release, and Operations.

RELATED CODEX SECTIONS

020 Systems Engineering; 039 Runtime Architecture; 228-230 Quality; 231 Software Testing; 240-242 Load, Stress, and Scalability Testing; 301-330 Operations.

CODEX/03_SECTIONS/239_PERFORMANCE_TESTING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, performance and operations context, and the integration roadmap. Exact workloads, objectives, environment, service levels, cost, instrumentation, and acceptance authority remain unresolved.

240

SECTION 240 - LOAD TESTING

CODEX/03_SECTIONS/240_LOAD_TESTING.txt

SHA-256 ae58a38c65bcf76fe6d1db90a0bedaed4f59772945ca1415d1f6a7eee66a1a6c

4,829 BYTES / PRESERVED SOURCE IDENTITY

SECTION 240 — Load Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Validate system behavior under expected and planned concurrent workload, including throughput, latency, resource use, queues, dependencies, and user-visible outcome.

REQUIRED OUTCOME

A load-test plan with workload model, ramp and duration, environment, data, service objectives, instrumentation, safety controls, results, bottlenecks, and capacity decision.

WHY THIS MATTERS

Expected load is not a single number. Arrival pattern, concurrency, data mix, cache state, dependency behavior, and duration determine whether a service remains useful or degrades.

APPLICABILITY

Apply when concurrent users, requests, jobs, devices, events, or transactions can affect capacity, latency, cost, or reliability.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no concurrent workload or capacity claim changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use workload, performance, architecture, capacity, cost, reliability, rate-limit, provider, data, and operations baselines. Establish authorization, isolation, and stop conditions.

REQUIRED INPUTS

Arrival and concurrency model, duration, data mix, users, dependencies, environment, limits, service objectives, metrics, baseline, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve workload, environment, cost, production-like testing, stop conditions, and capacity decision. Operators protect shared services and decide emergency stop.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft workload models, scripts, analysis, and bottleneck hypotheses. It must not launch unapproved load, use real sensitive users, overload dependencies, or claim capacity from a synthetic mismatch.

APPLICABLE EXISTING AGENT ROLES

Load Test Engineer, Performance Engineer, Tester, Architect, Analyst, Operations Engineer, Observer, Verifier, Quality Engineer, and Documentation Writer.

REQUIRED ACTIVITIES

Define expected load and success; prepare isolated environment and data; validate generator; ramp and sustain; monitor service and dependencies; capture resource and user outcome; stop safely; analyze; tune; rerun; and record capacity and limitations.

HUMAN-AI INTERACTION REQUIREMENTS

Show workload, generator, target, dependency, environment, rate, duration, metric, result, cost, and stop state. Require human approval before execution and tuning.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide rate limiting, backpressure, queue visibility, idempotency, autoscaling, resource isolation, dependency protection, synthetic identity, and cleanup.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show active load, rate, concurrency, duration, target, dependency health, metrics, cost, safety state, and authorized pause, stop, scale, rollback, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Load model; generator and data record; environment and authorization; telemetry; results; bottleneck and capacity analysis; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify workload, rate, concurrency, duration, version, environment, data, dependency, metric, method, result, and limitation. One load profile proves only that profile.

VERIFICATION AND ACCEPTANCE TESTS

Verify model realism, generator accuracy, safety, service objectives, queues, resource, dependency impact, autoscaling, degradation, recovery, cost, and repeatability.

FAILURE STATES AND ANTI-PATTERNS

Load against shared production without authority, generator bottleneck, unrealistic traffic, ignored dependency impact, throughput-only success, and load test that becomes an outage.

RECOVERY AND RECONSIDERATION

Stop load, protect dependencies, restore resources, preserve telemetry, and reassess capacity. Reopen on workload, architecture, provider, configuration, or objective change.

COMPLETION CRITERIA

Workload, authorization, environment, results, service objectives, safety, bottlenecks, capacity, evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve workload, generator, environment, telemetry, results, tuning, decisions, evidence, exceptions, and open gaps. Hand off to Performance, Stress, Scalability, Reliability, Release, and Operations.

RELATED CODEX SECTIONS

239 Performance Testing; 241 Stress Testing; 242 Scalability Testing; 301-330 Capacity, Availability, Performance, Reliability, and Operations.

CODEX/03_SECTIONS/240_LOAD_TESTING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, performance and operations context, and the integration roadmap. Exact load profile, service levels, environment, dependency limits, cost, and acceptance authority remain unresolved.

241

SECTION 241 - STRESS TESTING

CODEX/03_SECTIONS/241_STRESS_TESTING.txt

SHA-256 ccffcf883b92f1481cdb5987ac4968e1fde83f9a81e942b66566a0d57717896b

5,143 BYTES / PRESERVED SOURCE IDENTITY

SECTION 241 — Stress Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Evaluate behavior beyond expected operating conditions and determine how the system degrades, protects people and data, recovers, and returns to service.

REQUIRED OUTCOME

A stress-test model with overload or adverse condition, safety boundaries, observable thresholds, degradation expectations, recovery, resource exhaustion behavior, evidence, and owner decision.

WHY THIS MATTERS

Systems often fail in the transition between healthy and unavailable. Stress testing reveals queue collapse, retry storms, memory or resource exhaustion, unsafe fallback, data corruption, and recovery that was never exercised.

APPLICABILITY

Apply when overload, dependency loss, resource exhaustion, failure combination, extreme input, or environmental boundary can affect safety, continuity, integrity, or service.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no adverse condition or recovery claim is material and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use performance, load, reliability, resilience, safety, security, recovery, capacity, and operations baselines. Establish isolation, monitoring, stop, restore, and communication authority.

REQUIRED INPUTS

Stress scenario, limit, trigger, workload, environment, data, dependencies, safety constraints, expected degradation, recovery objective, instrumentation, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve adverse conditions, safety, data, production-like scope, stop, recovery, and residual risk. Operators control the exercise and can halt it immediately.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft stress scenarios, analyze telemetry, and identify degradation patterns. It must not create unsafe conditions, continue after stop, alter recovery, or claim resilience from a simulation without human authority and evidence.

APPLICABLE EXISTING AGENT ROLES

Resilience Engineer, Performance Engineer, Tester, Safety Reviewer, Architect, Analyst, Observer, Operations Engineer, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define adverse condition and safe bounds; prepare isolation and restore; establish baseline; apply controlled stress; monitor safety, integrity, resources, and users; stop; recover; analyze degradation and causes; remediate; rerun; record limits.

HUMAN-AI INTERACTION REQUIREMENTS

Show scenario, trigger, rate, target, safety state, evidence, expected degradation, actual impact, and stop control. Require human supervision throughout stress and recovery.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Design overload shedding, backpressure, circuit breaking, isolation, graceful degradation, data integrity, checkpointing, restore, failover, and operator visibility. Prevent retry or recovery mechanisms from amplifying harm.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show stress phase, condition, resources, safety, user impact, recovery state, evidence freshness, and authorized pause, stop, isolate, restore, scale, rollback, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Stress charter; scenario and safety plan; environment and data record; telemetry; degradation and recovery results; remediation; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify condition, trigger, workload, version, environment, resource, time, method, result, user impact, recovery, and limitation. A successful recovery in one failure mode proves only that mode.

VERIFICATION AND ACCEPTANCE TESTS

Verify stop, isolation, degradation, data integrity, alerting, operator response, recovery, restore, failover, cleanup, and return-to-service. Reproduce material results independently.

FAILURE STATES AND ANTI-PATTERNS

Stress without isolation, no stop path, destructive load, retry amplification, recovery not observed, data corruption ignored, and resilience claimed from uptime before failure.

RECOVERY AND RECONSIDERATION

Stop and stabilize, preserve evidence, restore or fail over under authority, protect users, communicate, and reassess residual risk. Reopen on architecture, workload, dependency, recovery, or safety change.

COMPLETION CRITERIA

Condition, safety, workload, degradation, recovery, evidence, owner, limits, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve stress and recovery traces, actions, findings, decisions, evidence, exceptions, and open gaps. Hand off to Reliability, Resilience, Capacity, Operations, Incident Response, and Testing.

RELATED CODEX SECTIONS

018 Risk Management; 029 Operational Architecture; 239-240 Performance and Load Testing; 242 Scalability Testing; 307-330 Reliability, Resilience, Continuity, and Recovery.

CODEX/03_SECTIONS/241_STRESS_TESTING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, resilience and testing context, and the integration roadmap. Exact stress conditions, safety, environment, recovery, service objectives, and acceptance authority remain unresolved.

242

SECTION 242 - SCALABILITY TESTING

CODEX/03_SECTIONS/242_SCALABILITY_TESTING.txt

SHA-256 9e6148fa9838450c4eb5609da406d84eadcf6095a1378385afbfa54e03267f21

5,509 BYTES / PRESERVED SOURCE IDENTITY

SECTION 242 — Scalability Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Determine whether a system can grow across users, data, traffic, devices, tenants, geography, workload, or time while preserving protected outcomes and acceptable cost.

REQUIRED OUTCOME

A scalability model with growth dimensions, workload projection, architecture assumptions, scale mechanism, limits, service objectives, cost, data behavior, operational burden, and evidence.

WHY THIS MATTERS

A system may handle more traffic but fail with data growth, tenant count, geographic spread, coordination, cost, or operator load. Linear extrapolation from a small test can hide thresholds and discontinuities.

APPLICABILITY

Apply when expected or strategic growth can affect performance, reliability, cost, privacy, security, data lifecycle, operations, or user experience.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no growth dimension or scale-dependent outcome changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use architecture, workload, performance, load, stress, capacity, cost, data, tenancy, operations, reliability, and roadmap baselines. Define growth horizon, assumptions, and decision owner.

REQUIRED INPUTS

Growth dimensions, projections, workload, data volume, tenants, geography, dependencies, architecture, scaling mechanism, limits, service objectives, cost, staffing, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners set growth assumptions, service and cost tradeoffs, investment, risk, and acceptance. Architects and operators validate capacity and operational consequences; humans decide whether extrapolation is sufficient.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized measurements, model growth, identify bottlenecks, and propose experiments or architecture options. It must expose assumptions and uncertainty and must not promise scale, trigger capacity changes, or optimize cost at the expense of protected outcomes without human review.

APPLICABLE EXISTING AGENT ROLES

Scalability Engineer, Architect, Performance Engineer, Data Engineer, Analyst, Researcher, Tester, Verifier, Optimizer, Operations Engineer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define growth dimensions and horizon; model workloads and data; baseline current behavior; test scale steps; measure service, cost, resources, data, security, and operator load; identify thresholds; plan capacity; validate; and record limits.

HUMAN-AI INTERACTION REQUIREMENTS

Show projection, assumption, measurement, environment, scale step, service objective, cost, uncertainty, and limitation. Require human approval for extrapolation, resource changes, customer commitments, and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support partitioning, sharding, caching, queues, autoscaling, data lifecycle, tenant isolation, rate limits, observability, migration, rollback, and operational staffing. Identify nonlinear thresholds and shared failure domains.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show growth dimension, current and projected load, threshold, resource, cost, service state, evidence freshness, owner, and authorized scale, throttle, defer, redesign, rollback, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Scalability charter; growth and workload model; scale-test plan; measurements and thresholds; capacity and cost analysis; architecture decisions; risk, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify growth dimension, version, workload, data, environment, scale step, metric, cost, method, result, uncertainty, and limitation. Extrapolation must state its assumptions and validation boundary.

VERIFICATION AND ACCEPTANCE TESTS

Verify workload and growth model, scale mechanism, thresholds, service objectives, data integrity, tenant isolation, cost, operator load, failure, recovery, and repeatability. Test beyond expected and boundary scale safely.

FAILURE STATES AND ANTI-PATTERNS

Linear extrapolation, traffic-only scaling, data and operator load ignored, cost surprise, shared tenant failure, autoscaling without guardrails, and growth commitments without evidence.

RECOVERY AND RECONSIDERATION

Throttle or isolate growth, preserve measurements, use approved capacity or fallback, protect users and data, and revisit architecture or assumptions. Reopen on growth, workload, data, provider, cost, or service-objective change.

COMPLETION CRITERIA

Growth, assumptions, workloads, mechanisms, limits, service, cost, owners, evidence, tests, recovery, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve projections, workloads, environments, measurements, thresholds, decisions, costs, evidence, exceptions, and open gaps. Hand off to Capacity, Performance, Load, Stress, Data, Reliability, Architecture, and Operations.

RELATED CODEX SECTIONS

020 Systems Engineering; 029 Operational Architecture; 031 Data Architecture; 239-241 Performance, Load, and Stress Testing; 301-330 Capacity, Performance, Reliability, and Operations.

CODEX/03_SECTIONS/242_SCALABILITY_TESTING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, architecture, performance, and operations context, and the integration roadmap. Exact growth, workload, horizon, service objectives, cost, scale mechanism, and acceptance authority remain unresolved.

243

SECTION 243 - RELIABILITY TESTING

CODEX/03_SECTIONS/243_RELIABILITY_TESTING.txt

SHA-256 15fa66a9b977cc02b809cbdc213f13335dae1de26aec1aa2aee6aa6565b5336b

5,275 BYTES / PRESERVED SOURCE IDENTITY

SECTION 243 — Reliability Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Measure whether a system performs its intended function consistently over time and under declared operating conditions, including faults, maintenance, recovery, and change.

REQUIRED OUTCOME

A reliability-test model with failure definition, workload, duration, environment, fault assumptions, service objectives, data integrity, repair or recovery behavior, evidence, and limits.

WHY THIS MATTERS

Reliability is time and condition dependent. A short successful test can miss wear, memory growth, intermittent faults, retry amplification, operator error, dependency loss, or recovery that does not restore the intended outcome.

APPLICABILITY

Apply when continuity, failure frequency, durability, correctness over time, maintenance, or recovery is material to people, service, equipment, data, or trust.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no reliability or continuity claim changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use requirements, architecture, operational, resilience, performance, failure, maintenance, backup, restore, and service-level baselines. Define test authority, duration, environment, and stop conditions.

REQUIRED INPUTS

Reliability objective, failure definition, workload, duration, unit or service, environment, faults, maintenance, telemetry, repair, recovery, prior incidents, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define reliability and recovery outcomes; operators protect the environment; testers document faults and exposure; humans accept residual failure risk and service tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized failure and telemetry records, draft scenarios, identify trends, and propose experiments. It must not invent reliability statistics, continue unsafe tests, suppress intermittent failures, or declare a service reliable from uptime alone.

APPLICABLE EXISTING AGENT ROLES

Reliability Engineer, Tester, Resilience Engineer, Architect, Operations Engineer, Analyst, Observer, Verifier, Quality Engineer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define objective and failure; select representative workload and duration; establish baseline; inject or observe faults; measure function, errors, repair, recovery, and data; analyze trends; remediate; rerun; and document confidence and limits.

HUMAN-AI INTERACTION REQUIREMENTS

Show unit or service, version, workload, duration, fault, environment, metric, recovery, result, uncertainty, and limitation. Require human approval for fault injection, production-like testing, and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide health signals, durable state, checkpointing, retry limits, graceful degradation, repair, failover, restore, maintenance, and evidence across long runs. Avoid recovery loops that hide failure or corrupt state.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show reliability objective, elapsed run, failures, fault state, recovery, service impact, evidence freshness, owner, and authorized pause, stop, repair, failover, restore, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Reliability charter; workload and fault model; environment and telemetry; run results; failure and recovery record; trend and confidence analysis; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify unit or service, version, workload, duration, condition, fault, method, time, result, recovery, sample, uncertainty, and limitation. Uptime or one long run proves only its observed conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify failure definition, representative workload, duration, telemetry, repair, recovery, data integrity, maintenance, repeatability, and confidence. Compare with incidents and field evidence.

FAILURE STATES AND ANTI-PATTERNS

Uptime as reliability, short test, healthy-only workload, hidden intermittent failures, recovery not counted, data loss ignored, and statistics without exposure or sample context.

RECOVERY AND RECONSIDERATION

Stabilize service, preserve evidence, repair or fail over, restore state, and reassess user and data impact. Reopen on failure pattern, workload, dependency, version, maintenance, or recovery change.

COMPLETION CRITERIA

Objective, workload, duration, faults, recovery, evidence, result, confidence, owner, limits, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve runs, telemetry, failures, repairs, recovery, decisions, evidence, exceptions, and open gaps. Hand off to Resilience, Chaos, Capacity, Operations, Incident Response, and Quality.

RELATED CODEX SECTIONS

029 Operational Architecture; 228-230 Quality; 239-242 Performance, Load, Stress, and Scalability; 244-245 Resilience and Chaos; 301-330 Reliability and Recovery.

CODEX/03_SECTIONS/243_RELIABILITY_TESTING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, reliability and operations context, and the integration roadmap. Exact failure model, duration, workload, service objective, recovery, field evidence, and acceptance authority remain unresolved.

244

SECTION 244 - RESILIENCE TESTING

CODEX/03_SECTIONS/244_RESILIENCE_TESTING.txt

SHA-256 e2f7fc6bdd3a391472824a0ba6debbd1d1dbaaf8b54215f223f8c27d66447dcc

5,209 BYTES / PRESERVED SOURCE IDENTITY

SECTION 244 — Resilience Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Validate the ability of a system, organization, or service to absorb disruption, adapt, preserve protected outcomes, recover, and learn without unsafe or unauthorized behavior.

REQUIRED OUTCOME

A resilience-test model with disruption scenarios, protected outcomes, dependencies, degradation, continuity, recovery, communications, evidence, owners, and residual-risk decision.

WHY THIS MATTERS

Resilience is more than availability. A service may stay online while losing integrity, privacy, safety, user agency, or recoverability. Recovery plans without exercised authority and evidence are assumptions.

APPLICABILITY

Apply when disruption, dependency loss, resource failure, human absence, attack, environmental change, or organizational shock can affect essential outcomes.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no protected outcome, dependency, continuity, recovery, or disruption claim is material and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use operational, continuity, disaster recovery, reliability, safety, security, capacity, backup, restore, communication, staffing, and ownership baselines.

REQUIRED INPUTS

Protected outcomes, disruption scenarios, dependencies, modes, service levels, recovery objectives, staffing, facilities, data, communications, runbooks, evidence, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define essential outcomes and acceptable degradation; commanders control exercise and recovery; humans approve service reduction, failover, communication, risk, and return to service.

AI RESPONSIBILITIES AND BOUNDARIES

AI may model dependencies, draft scenarios, analyze recovery evidence, and identify gaps. It must not declare resilience, trigger failover, change service mode, or suppress user and operator impact without delegated human authority.

APPLICABLE EXISTING AGENT ROLES

Resilience Engineer, Architect, Continuity Planner, Operations Engineer, Tester, Observer, Analyst, Verifier, Safety Reviewer, Security Auditor, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define outcomes and modes; map dependencies and common failure; select controlled disruptions; establish safety and stop; execute; observe technical and human response; recover; communicate; assess residual impact; remediate; and repeat.

HUMAN-AI INTERACTION REQUIREMENTS

Show disruption, outcome, dependency, authority, mode, evidence, user impact, recovery, and stop state. Require human control for exercise, failover, service reduction, communication, and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide redundancy, isolation, graceful degradation, data integrity, fallback, manual operation, checkpoint, backup, restore, failover, communication, and control-plane independence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show current mode, disruption, protected outcome, dependency, recovery objective, operator, evidence freshness, and authorized pause, degrade, fail over, restore, communicate, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Resilience charter; scenario and outcome map; dependency and recovery model; exercise and safety plan; results; communication and lessons record; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify outcome, scenario, version, environment, dependency, authority, method, time, result, impact, recovery, and limitation. A failover drill proves only the exercised path and conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify disruption control, safe degradation, integrity, operator response, staffing, communication, failover, restore, data reconciliation, return to service, and lesson closure.

FAILURE STATES AND ANTI-PATTERNS

Availability-only resilience, exercise without stop, dependency correlation ignored, recovery dependent on failed system, user impact hidden, and drill success treated as durable resilience.

RECOVERY AND RECONSIDERATION

Stabilize, preserve evidence, operate in approved degraded mode, recover or fail over, communicate, and reassess protected outcomes. Reopen on dependency, threat, staffing, architecture, or recovery change.

COMPLETION CRITERIA

Outcomes, scenarios, dependencies, modes, recovery, owners, evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve scenarios, results, recovery, communications, decisions, evidence, exceptions, and open gaps. Hand off to Chaos, Reliability, Disaster Recovery, Operations, Security, Safety, and Governance.

RELATED CODEX SECTIONS

005 Continuity and Recovery; 029 Operational Architecture; 244 Resilience; 245 Chaos Engineering; 301-330 Reliability, Continuity, Disaster Recovery, Backup, Restore, and Failover.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, operational and continuity context, and the integration roadmap. Exact protected outcomes, scenarios, recovery objectives, staffing, communications, and acceptance authority remain unresolved.

245

SECTION 245 - CHAOS ENGINEERING

CODEX/03_SECTIONS/245_CHAOS_ENGINEERING.txt

SHA-256 46036afcc3f127bb208491d2e792014a9a754b47b22b3d3deed2f2d410eb52e2

5,178 BYTES / PRESERVED SOURCE IDENTITY

SECTION 245 — Chaos Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Learn how systems and organizations behave under controlled, hypothesis-driven disruption and use the evidence to improve resilience, detection, recovery, and human control.

REQUIRED OUTCOME

A governed chaos experiment with steady-state hypothesis, scope, authorization, blast radius, safety, abort, observability, result, learning, remediation, and repeatability.

WHY THIS MATTERS

Resilience assumptions remain hidden until challenged. Uncontrolled chaos is simply an incident; controlled experiments require safety, observability, a meaningful hypothesis, and authority to stop.

APPLICABILITY

Apply when controlled disruption can safely answer a material resilience or operational question that ordinary testing cannot.

NON-APPLICABILITY CONDITIONS

Record non-applicability when risk or authority cannot support a controlled experiment, a lower-risk test answers the question, or no material resilience uncertainty exists.

PREREQUISITES AND DEPENDENCIES

Define steady state, hypothesis, scope, blast radius, authorization, safety, abort, contacts, observability, rollback, recovery, communication, and incident conversion.

REQUIRED INPUTS

Hypothesis, protected outcome, target, version, dependency, disruption, limits, environment, telemetry, rollback, responders, data, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Experiment owners approve hypothesis, scope, risk, safety, stop, communication, and recovery. A controller can abort immediately; humans interpret learning and decide remediation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft hypotheses, dependency maps, experiment plans, and analysis. It must not initiate or escalate disruption, conceal impact, change targets, or declare steady state without explicit human authority and live supervision.

APPLICABLE EXISTING AGENT ROLES

Chaos Engineer, Resilience Engineer, Architect, Operations Engineer, Tester, Observer, Analyst, Verifier, Safety Reviewer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define steady state and hypothesis; authorize and bound; verify observability and abort; run smallest safe experiment; observe technical and human outcome; abort or recover; analyze; remediate; repeat; and record learning.

HUMAN-AI INTERACTION REQUIREMENTS

Show hypothesis, target, disruption, blast radius, authority, steady state, evidence, impact, and abort. Require human presence for consequential experiments and immediate stop capability.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support safe injection, isolation, feature flags, rate limits, dependency substitution, rollback, recovery, telemetry, and incident conversion. Keep experiment controls separate from ordinary production authority.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show experiment phase, hypothesis, target, impact, steady state, abort state, evidence, owner, and authorized pause, abort, recover, narrow, remediate, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Experiment charter; hypothesis and steady-state definition; authorization and safety plan; run record; telemetry and impact; learning and remediation; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify experiment, hypothesis, target, version, disruption, environment, authority, time, observation, result, impact, recovery, and limitation. A successful experiment proves only its declared hypothesis and conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify authorization, steady state, observability, abort, blast radius, recovery, incident conversion, data integrity, operator response, and repeatability. Confirm no unapproved target or person was affected.

FAILURE STATES AND ANTI-PATTERNS

Chaos without hypothesis, no abort, broad blast radius, missing telemetry, surprise to responders, experiment mistaken for incident, and learning without remediation.

RECOVERY AND RECONSIDERATION

Abort, stabilize, preserve evidence, recover, communicate, and convert to incident handling when necessary. Reassess hypothesis and safeguards before repeating. Reopen on system, dependency, threat, or operational change.

COMPLETION CRITERIA

Hypothesis, scope, safety, authority, evidence, result, impact, recovery, learning, remediation, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve experiment identity, controls, timeline, telemetry, findings, recovery, decisions, evidence, exceptions, and open gaps. Hand off to Resilience, Reliability, Security Operations, Incident Response, Operations, and Governance.

RELATED CODEX SECTIONS

029 Operational Architecture; 206 Security Operations; 243 Reliability Testing; 244 Resilience Testing; 301-330 Reliability, Resilience, Continuity, Disaster Recovery, and Failover.

CODEX/03_SECTIONS/245_CHAOS_ENGINEERING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, resilience and operations context, and the integration roadmap. Exact experiment authority, steady state, blast radius, safety, environment, and acceptance evidence remain unresolved.

246

SECTION 246 - SECURITY TESTING

CODEX/03_SECTIONS/246_SECURITY_TESTING.txt

SHA-256 62e408e8397e0ec670ddd762c3da4b6cf745287d94c3020af5d5fdf0824e7c6f

5,115 BYTES / PRESERVED SOURCE IDENTITY

SECTION 246 — Security Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Evaluate whether security requirements, controls, trust boundaries, identities, data, interfaces, configurations, and recovery paths behave as intended under authorized conditions.

REQUIRED OUTCOME

A risk-based security-test plan and evidence package with scope, threat basis, authorization, methods, findings, severity, remediation, retest, limitations, and owner decision.

WHY THIS MATTERS

Security tests can be too narrow, unsafe, stale, or detached from actual threat and user context. A clean scan or passing control test proves only its method, version, coverage, and time.

APPLICABILITY

Apply when a material security property, trust boundary, identity, data, interface, dependency, configuration, or defensive control changes or requires assurance.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no security property or threat-relevant behavior changes and an approved exact baseline provides sufficient current evidence.

PREREQUISITES AND DEPENDENCIES

Use threat model, security architecture, IAM, cryptography, vulnerability, operations, privacy, test, incident, and authorization baselines. Define scope, handling, safety, and stop authority.

REQUIRED INPUTS

Threats, assets, requirements, identities, interfaces, data, versions, configurations, test accounts, methods, exclusions, evidence, findings, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Security owners authorize scope and methods, control sensitive data, interpret findings, approve remediation or exception, and accept residual risk. Testers stop on unsafe or out-of-scope behavior.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft test cases, analyze authorized results, identify control gaps, and propose remediation. It must not scan or exploit without authorization, expose secrets, broaden scope, suppress findings, or declare security from generated analysis.

APPLICABLE EXISTING AGENT ROLES

Security Tester, Security Auditor, Threat Modeler, Penetration Tester, Researcher, Analyst, Implementer, Verifier, Red Team, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Map threats to tests; confirm authority; prepare safe environment and data; test preventive, detective, responsive, and recovery controls; record evidence; triage; remediate; retest; report limits; and reopen on change.

HUMAN-AI INTERACTION REQUIREMENTS

Show threat, asset, version, method, authority, data, environment, result, severity, evidence, and limitation. Require human review for execution, high-impact findings, containment, disclosure, exception, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support isolated test identities, traceable configurations, safe fixtures, logging, rollback, fault injection, evidence protection, and separation from production authority.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show threat, scope, active test, findings, severity, evidence freshness, remediation, owner, and authorized start, pause, stop, remediate, accept, disclose, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Security-test strategy; threat-to-test matrix; authorization and rules; cases and data; results and findings; remediation and retest; exception, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, version, threat, method, authorization, environment, data, test, result, time, actor, impact, and limitation. Preserve sensitive evidence with least privilege.

VERIFICATION AND ACCEPTANCE TESTS

Verify authorization, scope, threat coverage, control behavior, logging, findings, remediation, retest, cleanup, privacy, and independent reproduction. Test false positives and missing visibility.

FAILURE STATES AND ANTI-PATTERNS

Unapproved testing, scanner-only assurance, scope drift, exploit or secret exposure, security test data in production, clean result treated as total security, and findings closed without retest.

RECOVERY AND RECONSIDERATION

Stop, preserve state, contain, notify, clean up, remediate, and reassess affected assets and users. Reopen on threat, version, configuration, dependency, incident, or evidence change.

COMPLETION CRITERIA

Threats, scope, authority, methods, results, findings, remediation, evidence, limitations, owners, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve authorization, cases, findings, evidence, remediation, retest, exceptions, and open gaps. Hand off to Vulnerability, Security Operations, Red Team, Blue Team, Incident Response, Release, and Governance.

RELATED CODEX SECTIONS

045 Security Architecture; 198-209 Security and Protection; 210-212 Penetration, Red, and Blue Teaming; 231-254 Testing; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, security and testing context, and the integration roadmap. Exact threats, methods, authorization, environments, data, standards, and acceptance authority remain unresolved.

247

SECTION 247 - ACCESSIBILITY TESTING

CODEX/03_SECTIONS/247_ACCESSIBILITY_TESTING.txt

SHA-256 6e95737b19e1362de5c77101bd90e017be9bd57b86d0c9fa16f1f1e94f15c0e2

5,709 BYTES / PRESERVED SOURCE IDENTITY

SECTION 247 — Accessibility Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Verify that people with varied abilities, devices, input methods, languages, and assistive technologies can perceive, understand, navigate, operate, and recover through the declared experience.

REQUIRED OUTCOME

An accessibility-test model with users, standards, journeys, assistive technology, environments, criteria, defects, accommodations, evidence, and owner decision.

WHY THIS MATTERS

Automated checks find some structural issues but cannot establish actual comprehension, operability, dignity, or recovery. Inaccessible design can deny access even when the feature is technically functional.

APPLICABILITY

Apply to user interfaces, documentation, devices, services, communication, support, physical environments, data presentations, and AI interactions intended for or affecting people.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no person-facing or human-control surface changes and an approved exact baseline covers the scope. Internal or “temporary” tools may still require accessibility when people rely on them.

PREREQUISITES AND DEPENDENCIES

Use user research, accessibility requirements, design, content, platform, localization, privacy, security, support, and legal baselines. Include disabled users or qualified specialists and assistive technology where relevant.

REQUIRED INPUTS

Users, abilities, journeys, content, controls, platform, browser or device, assistive technology, standards, language, data, errors, recovery, accommodations, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners fund and prioritize accessibility; qualified reviewers and users evaluate real experience; humans accept or reject exceptions and provide accommodation and remediation. Disabled participants must not be treated as test data without care and consent.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized markup, content, interaction traces, and automated results; draft test cases and alternatives; and identify likely barriers. It must not claim accessibility from static inspection, speak for disabled people, expose personal data, or approve an exception without human and user-informed review.

APPLICABLE EXISTING AGENT ROLES

Accessibility Reviewer, User Experience Reviewer, Researcher, Tester, Verifier, Content Designer, Analyst, Critic, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Define users and criteria; test keyboard, touch, voice, screen reader, magnification, contrast, motion, timing, language, cognitive load, error, support, and recovery; include disabled users where appropriate; record barriers; remediate; retest; and preserve limitations.

HUMAN-AI INTERACTION REQUIREMENTS

Show user, assistive technology, platform, journey, content, method, result, and limitation. Require human review and lived-experience input for acceptance. Provide accessible owner and appeal controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support semantic structure, keyboard and alternative input, focus, captions, text alternatives, reflow, contrast, timing control, error recovery, accessible authentication, localization, and non-visual owner control.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show accessibility scope, users, defects, severity, platform coverage, evidence freshness, accommodations, owner, and authorized prioritize, remediate, accept, provide alternative, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Accessibility charter; user and technology matrix; journey and criteria; manual and automated results; participant and accommodation record; defect, exception, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify user or specialist, platform, assistive technology, version, journey, method, result, barrier, impact, and limitation. An automated scan or conformance claim proves only its rule set and scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify keyboard, non-visual, touch, voice, magnification, contrast, captions, semantics, timing, errors, forms, authentication, language, support, and recovery. Reproduce with representative technology and users.

FAILURE STATES AND ANTI-PATTERNS

Accessibility by scan only, disabled people consulted too late, inaccessible test or owner controls, color-only meaning, time limits without extension, and exception accepted without alternative or expiry.

RECOVERY AND RECONSIDERATION

Provide accessible alternative or support, preserve user impact, remediate, and retest with affected users. Reopen on content, platform, assistive technology, interaction, policy, or user-population change.

COMPLETION CRITERIA

Users, platforms, technologies, journeys, criteria, findings, accommodations, evidence, owners, exceptions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending user-informed review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve criteria, participant care, platform results, defects, accommodations, decisions, evidence, exceptions, and open gaps. Hand off to UX, Engineering, Content, Localization, Support, Legal, Testing, and Operations.

RELATED CODEX SECTIONS

001 User Research; 019 Requirements; 048 Accessibility Architecture; 230 Quality Assurance; 236-237 E2E and Acceptance Testing; 249 Usability Testing; 250 Localization; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, accessibility architecture, user research, and the integration roadmap. Exact standard, users, assistive technology, jurisdiction, accommodations, content, and acceptance authority remain unresolved.

248

SECTION 248 - COMPATIBILITY TESTING

CODEX/03_SECTIONS/248_COMPATIBILITY_TESTING.txt

SHA-256 97a06a58922d74ab3e5244da7f34c8524deaf0ba9ca87a3d29ac1a92c9f1c0ca

5,059 BYTES / PRESERVED SOURCE IDENTITY

SECTION 248 — Compatibility Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Verify that a capability behaves acceptably across supported versions, platforms, devices, browsers, providers, configurations, interfaces, data formats, and neighboring systems.

REQUIRED OUTCOME

A compatibility matrix with supported combinations, assumptions, contracts, prioritized coverage, results, known incompatibilities, migration or fallback, and owner decision.

WHY THIS MATTERS

Compatibility failures are often conditional on version, configuration, locale, hardware, provider, or data. A successful test on one combination cannot justify a universal support claim.

APPLICABILITY

Apply when users, systems, devices, providers, formats, environments, or versions must interoperate or remain supported across change.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no supported combination, interface, version, or consumer behavior changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use support policy, architecture, contracts, version management, user population, platform, localization, data, release, and operations baselines.

REQUIRED INPUTS

Supported matrix, versions, configurations, devices, providers, interfaces, data formats, users, priorities, known limits, test environments, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define support and deprecation; testers prioritize combinations; domain and support owners judge user impact. Humans approve unsupported behavior, migration, fallback, and communication.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze compatibility metadata, propose matrix coverage, generate cases, and summarize failures. It must not infer support from similarity, silently drop combinations, or make compatibility commitments without human review.

APPLICABLE EXISTING AGENT ROLES

Compatibility Tester, Architect, Release Engineer, Researcher, Analyst, Tester, Verifier, User Experience Reviewer, Support Reviewer, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define support; inventory combinations; prioritize risk and population; prepare environments and data; execute functional, interface, performance, accessibility, security, and migration cases; record incompatibility; provide fallback or communication; retest.

HUMAN-AI INTERACTION REQUIREMENTS

Show combination, support status, version, configuration, user, data, method, result, and limitation. Require human approval for support, deprecation, fallback, and customer communication.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use explicit compatibility contracts, feature detection, version negotiation, migration, graceful fallback, observability, configuration capture, and support boundaries. Avoid silent behavior changes.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show matrix, tested combinations, failures, support status, evidence freshness, owner, fallback, deprecation, and authorized support, restrict, migrate, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Compatibility matrix; support and deprecation policy; environment record; results and incompatibility register; migration and fallback plan; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify combination, version, configuration, device or provider, data, user, method, result, time, and limitation. A passing combination proves only that combination.

VERIFICATION AND ACCEPTANCE TESTS

Verify prioritized coverage, interfaces, data, performance, security, accessibility, migration, fallback, documentation, and support. Reproduce material incompatibilities and confirm boundaries.

FAILURE STATES AND ANTI-PATTERNS

Support by assumption, matrix explosion without prioritization, silent drop of old users, provider behavior assumed identical, fallback untested, and compatibility claim from one platform.

RECOVERY AND RECONSIDERATION

Restrict unsupported combinations, provide fallback or migration, communicate, preserve failure evidence, and reassess user impact. Reopen on platform, provider, version, data, or support change.

COMPLETION CRITERIA

Support, combinations, cases, results, incompatibilities, fallback, owners, evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve matrix and versions, results, support decisions, migrations, exceptions, evidence, and open gaps. Hand off to Release, Version Management, Deprecation, Support, Operations, UX, and Testing.

RELATED CODEX SECTIONS

019 Requirements; 020 Systems Engineering; 248 Compatibility Testing; 388 Version Management; 389 Deprecation Management; 378-381 Updates; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, support and version context, and the integration roadmap. Exact matrix, support population, platforms, providers, migration, deprecation, and acceptance authority remain unresolved.

249

SECTION 249 - USABILITY TESTING

CODEX/03_SECTIONS/249_USABILITY_TESTING.txt

SHA-256 fe826de1fa7bd0fcc66f2e3e4c8aad6cd188af87ee223bb5ad561e0191478de7

5,099 BYTES / PRESERVED SOURCE IDENTITY

SECTION 249 — Usability Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Learn whether representative people can understand, operate, recover from, and derive intended value from a capability in realistic context.

REQUIRED OUTCOME

A usability-test model with users, tasks, context, prototype or product, observation method, consent, accessibility, findings, severity, design response, and limitations.

WHY THIS MATTERS

Usability is contextual and cannot be inferred from interface polish, completion analytics, or expert opinion alone. A task can be completed while users misunderstand risk, lose agency, or need hidden support.

APPLICABILITY

Apply when a person must understand, decide, operate, recover, learn, or trust a user-facing or operator-facing capability.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no human-facing behavior, task, decision, support, or control changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use user research, requirements, design, accessibility, privacy, safety, security, localization, support, and acceptance baselines. Obtain informed participation and protect participant data.

REQUIRED INPUTS

Users, tasks, context, goals, content, prototype or product, hypotheses, success criteria, accessibility, consent, data handling, observer, and acceptance questions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Researchers recruit and protect participants; owners accept findings and prioritize change; humans interpret lived experience and decide tradeoffs. Participants may stop, withdraw, or report harm.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft tasks, analyze authorized notes, identify patterns, and propose design questions. It must not fabricate users, infer intent from sparse behavior, expose participant data, or decide usability or release without human interpretation.

APPLICABLE EXISTING AGENT ROLES

User Researcher, User Experience Reviewer, Accessibility Reviewer, Researcher, Analyst, Observer, Tester, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define question and users; recruit ethically; prepare realistic task and context; observe comprehension, behavior, errors, confidence, and recovery; analyze; share findings; design changes; retest; and preserve participant care and limitations.

HUMAN-AI INTERACTION REQUIREMENTS

Show participant context, task, observation, interpretation, evidence, uncertainty, and privacy handling. Require human review and do not represent AI synthesis as direct participant voice.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support accessible interaction, explainability, undo, error recovery, support, privacy, instrumentation, safe prototypes, and separation of participant data from production records.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show research question, users, tasks, findings, severity, evidence freshness, participant care, owner, and authorized prioritize, redesign, accommodate, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Usability charter; participant and consent record; task and context set; observation and finding ledger; design response; test, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify participant or segment, task, context, method, observer, time, observation, interpretation, impact, confidence, and limitation. Completion rate or quote alone proves only its bounded claim.

VERIFICATION AND ACCEPTANCE TESTS

Verify participant fit, consent, task realism, accessibility, comprehension, error recovery, privacy, analysis traceability, design response, and repeatability. Include people who use support or alternative methods.

FAILURE STATES AND ANTI-PATTERNS

Testing friends or ideal users only, leading tasks, treating analytics as understanding, ignoring failure and recovery, exposing participant data, and optimizing speed over comprehension or safety.

RECOVERY AND RECONSIDERATION

Protect participants, stop harmful study, correct design, preserve findings, and retest with affected users. Reopen on population, context, task, interface, content, or risk change.

COMPLETION CRITERIA

Question, users, tasks, context, evidence, findings, limitations, design response, owners, participant care, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending user-informed review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve consent, participant protections, findings, design decisions, evidence, exceptions, and open gaps. Hand off to Product, UX, Accessibility, Engineering, Support, Acceptance, Privacy, and Operations.

RELATED CODEX SECTIONS

001 User Research; 019 Requirements; 048 Accessibility Architecture; 237 Acceptance Testing; 247 Accessibility Testing; 250 Localization; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, user research, accessibility, and testing context, and the integration roadmap. Exact users, tasks, participation, language, context, privacy, and acceptance authority remain unresolved.

250

SECTION 250 - LOCALIZATION TESTING

CODEX/03_SECTIONS/250_LOCALIZATION_TESTING.txt

SHA-256 ceab88aac0d2442881ee6af0027af6a0d78ae46698bb0652ec6eb6fab460c50a

5,272 BYTES / PRESERVED SOURCE IDENTITY

SECTION 250 — Localization Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Verify that a capability remains understandable, functional, respectful, accessible, and safe across declared languages, regions, scripts, formats, cultural contexts, and legal settings.

REQUIRED OUTCOME

A localization-test model with locales, content, translators, formats, layout, input, data, legal or cultural constraints, accessibility, defects, evidence, and owner decision.

WHY THIS MATTERS

Translation can preserve words while breaking meaning, layout, dates, numbers, consent, safety, humor, or workflow. Locale support is not proven by string coverage alone.

APPLICABILITY

Apply when users, content, data, interfaces, communications, support, or obligations vary by language, locale, script, region, calendar, currency, or cultural context.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no user-facing content, locale, format, legal context, or regional behavior changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use user research, content, accessibility, legal, privacy, data, design, platform, support, and release baselines. Identify locale owners, qualified reviewers, and sensitive terminology.

REQUIRED INPUTS

Locales, strings, content, terminology, formats, data, workflows, translators, devices, standards, legal notices, accessibility, test cases, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Locale and domain owners approve meaning, terminology, legal text, cultural risk, and acceptance. Qualified native or contextual reviewers identify harm, ambiguity, and exclusion; humans decide regional release.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft translations, identify missing strings, compare terminology, and generate locale test cases. It must not replace qualified review, invent legal or cultural meaning, expose private content, or publish localized material without human approval.

APPLICABLE EXISTING AGENT ROLES

Localization Reviewer, User Experience Reviewer, Accessibility Reviewer, Researcher, Content Designer, Tester, Verifier, Legal Advisor, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Inventory locale surfaces; define terminology; localize and review; test layout, input, sorting, dates, numbers, currency, time, plural, script, accessibility, legal, and support; record defects; remediate; retest; and monitor feedback.

HUMAN-AI INTERACTION REQUIREMENTS

Show locale, source, translator or model, terminology, context, legal status, reviewer, result, and limitation. Require human native or domain review for sensitive, legal, safety, or culturally consequential content.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Externalize content, support locale-aware formats, plural and directionality, fallback, Unicode, collation, input, accessibility, versioning, and safe handling of untranslated or stale content.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show locale coverage, content version, missing or stale strings, defects, reviewer, evidence freshness, owner, and authorized hold, correct, fallback, publish, restrict, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Locale matrix; terminology and content register; translator and reviewer record; locale test results; legal and accessibility findings; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify locale, content version, source, context, format, platform, reviewer, method, result, impact, and limitation. String completion or machine translation score proves only its bounded metric.

VERIFICATION AND ACCEPTANCE TESTS

Verify meaning, layout, direction, input, plural, date, number, currency, time, sorting, search, accessibility, legal, safety, support, and fallback. Test missing, long, malformed, and mixed-language content.

FAILURE STATES AND ANTI-PATTERNS

Machine translation without review, untranslated critical text, overflow, wrong units or dates, culturally harmful wording, legal notice drift, and locale pass based only on string count.

RECOVERY AND RECONSIDERATION

Hold or roll back affected locale, provide safe fallback, correct content, notify users where needed, and preserve findings. Reopen on content, locale, law, product, platform, or terminology change.

COMPLETION CRITERIA

Locales, content, terminology, reviewers, formats, accessibility, legal, results, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve locale and content versions, terminology, reviews, defects, decisions, evidence, exceptions, and open gaps. Hand off to UX, Content, Accessibility, Legal, Release, Support, Testing, and Operations.

RELATED CODEX SECTIONS

001 User Research; 019 Requirements; 247 Accessibility Testing; 249 Usability Testing; 251 Compliance Testing; 237 Acceptance Testing; 388 Version Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, user, content, accessibility, and testing context, and the integration roadmap. Exact locales, reviewers, legal text, terminology, standards, and acceptance authority remain unresolved.

251

SECTION 251 - COMPLIANCE TESTING

CODEX/03_SECTIONS/251_COMPLIANCE_TESTING.txt

SHA-256 8b3c3f77da96a1beb16fee228191c151b4c257ce9d6849a1c638b748260cbd05

5,149 BYTES / PRESERVED SOURCE IDENTITY

SECTION 251 — Compliance Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Test whether declared controls and operating practices satisfy applicable legal, regulatory, contractual, policy, or standard-based criteria for their exact scope and period.

REQUIRED OUTCOME

A compliance-test plan with authority, criteria, scope, sample, evidence, test method, exceptions, findings, remediation, attestation boundaries, and qualified owner decision.

WHY THIS MATTERS

Compliance is contextual and time-bound. A control can be designed but not operating, a certificate can be stale, and an automated check can miss the obligation's meaning or population.

APPLICABILITY

Apply when a claim of compliance, conformity, certification, attestation, contractual adherence, or regulatory control is required or material.

NON-APPLICABILITY CONDITIONS

Record non-applicability only after qualified review of authority, jurisdiction, entity, activity, period, and obligation, with the alternative evidence and reopening condition recorded.

PREREQUISITES AND DEPENDENCIES

Use legal, regulatory, policy, records, control, audit, privacy, security, quality, and evidence baselines. Define qualified reviewer, criteria, and attestation authority.

REQUIRED INPUTS

Authority, criteria, scope, obligations, controls, population, period, evidence, samples, owners, exceptions, incidents, prior findings, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Qualified compliance, legal, audit, or domain owners approve criteria, interpretation, samples, findings, reports, exceptions, and attestations. Humans never sign beyond evidence or delegate legal judgment to AI.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize criteria, map controls, sample authorized evidence, identify omissions, and draft workpapers. It must preserve source, authority, period, sample, and uncertainty; it must not declare compliance, sign, conceal findings, or communicate externally without human approval.

APPLICABLE EXISTING AGENT ROLES

Compliance Tester, Legal Advisor, Auditor, Governance Reviewer, Security Auditor, Privacy Reviewer, Researcher, Analyst, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Confirm authority and scope; define criteria and samples; collect evidence; test design and operation; record exceptions; validate with owners; report; remediate; follow up; preserve workpapers; and monitor change.

HUMAN-AI INTERACTION REQUIREMENTS

Show authority, criterion, scope, period, sample, control, evidence, method, result, limitation, and reviewer. Require qualified human acceptance for conclusion, exception, attestation, and external report.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link authority, obligation, control, population, evidence, finding, response, and report. Preserve immutable workpapers, access restrictions, legal holds, segregation, and source versions.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show authority, scope, criteria, evidence coverage, findings, exceptions, due dates, reviewer, attestation status, and authorized assign, challenge, remediate, attest, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Compliance-test charter; criteria and sample; workpapers; evidence register; findings and exception record; remediation and follow-up; attestation boundary; decision and continuity records.

EVIDENCE REQUIREMENTS

Identify authority, criterion, scope, period, sample, control, source, method, result, reviewer, and limitation. A clean test proves only its criteria, sample, period, and method.

VERIFICATION AND ACCEPTANCE TESTS

Verify applicability, source currency, sample integrity, control operation, evidence, exceptions, remediation, report, attestation, and change monitoring. Obtain qualified sign-off.

FAILURE STATES AND ANTI-PATTERNS

Checklist compliance, unsupported attestation, stale criteria, sample manipulation, control description as evidence, and AI-generated report treated as qualified assurance.

RECOVERY AND RECONSIDERATION

Withdraw unsupported result, preserve workpapers, notify authority, remediate, amend report, and reassess deadline or disclosure. Reopen on obligation, scope, period, control, or evidence change.

COMPLETION CRITERIA

Authority, criteria, scope, sample, evidence, results, findings, remediation, qualification, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve criteria, workpapers, evidence, findings, responses, attestations, decisions, exceptions, and open gaps. Hand off to Legal, Compliance, Audit, Records, Controls, Security, Privacy, Quality, and Governance.

RELATED CODEX SECTIONS

219 Legal; 220 Regulatory Compliance; 221 Policy; 225 Records Management; 226 Audit; 227 Internal Controls; 228-230 Quality; 402-406 Review and Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, compliance and evidence context, and the integration roadmap. Exact authority, criteria, jurisdiction, period, sample, attestation, and acceptance requirements remain unresolved.

252

SECTION 252 - SAFETY TESTING

CODEX/03_SECTIONS/252_SAFETY_TESTING.txt

SHA-256 548fa4cc8f062193eff2ecd7ba453f8914fc1af50902006688e9511eb6391482

5,478 BYTES / PRESERVED SOURCE IDENTITY

SECTION 252 — Safety Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Verify that a system, product, process, or human-AI capability does not create unacceptable hazard under normal, foreseeable misuse, degraded, emergency, maintenance, and recovery conditions.

REQUIRED OUTCOME

A safety-test model with hazards, protected people, assumptions, operating modes, controls, test authority, safe limits, evidence, failures, recovery, and qualified acceptance.

WHY THIS MATTERS

Safety depends on interactions among technical behavior, humans, environment, procedures, maintenance, and failure. Simulation, certification, or ordinary operation cannot prove every hazardous condition is controlled.

APPLICABILITY

Apply when failure, misuse, automation, environment, or operation can cause injury, illness, unsafe condition, environmental harm, property damage, or loss of critical control.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when qualified safety analysis confirms no hazard or safety-relevant change for the exact scope, population, environment, and lifecycle.

PREREQUISITES AND DEPENDENCIES

Use safety architecture, hazard analysis, requirements, hardware, software, human factors, operations, maintenance, emergency, legal, and incident baselines. Assign qualified safety authority and stop control.

REQUIRED INPUTS

Hazards, people, environment, modes, controls, assumptions, failure conditions, interfaces, procedures, equipment, data, test limits, emergency response, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Qualified safety owners define hazards and limits, authorize testing, stop unsafe work, review evidence, and accept residual risk. Operators and affected people receive clear procedures, training, protection, and reporting paths.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized safety artifacts, enumerate scenarios, draft tests, and identify inconsistencies. It must not infer safety, control hazardous equipment, suppress warnings, approve operation, or substitute simulated confidence for qualified human judgment.

APPLICABLE EXISTING AGENT ROLES

Safety Engineer, Safety Reviewer, Test Engineer, Systems Engineer, Researcher, Observer, Analyst, Verifier, Critic, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Identify hazards and modes; define safe state and controls; authorize test; prepare protected environment and equipment; test normal, misuse, fault, emergency, maintenance, and recovery; record exposure; remediate; retest; and obtain qualified acceptance.

HUMAN-AI INTERACTION REQUIREMENTS

Show hazard, protected person, condition, authority, limit, control, result, evidence, and stop state. Require qualified human supervision for all safety-relevant execution and acceptance.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide independent safety functions, safe halt, interlocks, monitoring, redundancy, manual override, fault containment, clear alarms, recovery, and evidence independent of the failed component where required.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show hazard state, mode, safety control, affected people, test state, evidence freshness, owner, and authorized stop, isolate, safe-halt, recover, notify, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Safety-test charter; hazard and scenario set; authorization and protection plan; equipment and environment record; results and failures; recovery and acceptance; evidence, decision, and continuity records.

EVIDENCE REQUIREMENTS

Identify hazard, mode, unit, environment, condition, method, observer, time, result, exposure, recovery, and limitation. Certification or simulation proves only its declared standard and scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify hazard coverage, safe limits, controls, alarms, human response, fault containment, emergency, maintenance, recovery, accessibility, evidence, and qualified review. Stop on unexpected hazard.

FAILURE STATES AND ANTI-PATTERNS

Testing without safety authority, unsafe setup, alarm ignored, safe state unverified, human error excluded, simulation treated as field proof, and residual hazard hidden as an exception.

RECOVERY AND RECONSIDERATION

Stop, protect people, enter safe state, preserve evidence, provide emergency response, notify authority, and investigate. Do not resume until qualified review confirms controls and residual risk. Reopen on hazard, mode, environment, population, or design change.

COMPLETION CRITERIA

Hazards, people, modes, controls, limits, authority, results, recovery, evidence, residual risk, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified safety acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve hazard and test records, safety decisions, incidents, evidence, exceptions, training, recovery, and open gaps. Hand off to Safety, Hardware, Embedded, Operations, Incident Response, Quality, Legal, and Governance.

RELATED CODEX SECTIONS

050 Safety Architecture; 198-199 Embedded and Hardware Security; 232 Hardware Testing; 243-245 Reliability, Resilience, and Chaos; 301-330 Operations and Recovery; 402-404 Review.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, safety and testing context, and the integration roadmap. Exact hazard standard, qualified authority, environment, population, emergency method, jurisdiction, and acceptance evidence remain unresolved.

253

SECTION 253 - MODEL TESTING

CODEX/03_SECTIONS/253_MODEL_TESTING.txt

SHA-256 8995c53721c963b067bb7aa76b1833e7817407ed459e36fec8f1ced928fdf409

5,586 BYTES / PRESERVED SOURCE IDENTITY

SECTION 253 — Model Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Evaluate model behavior, performance, robustness, fairness, calibration, safety, privacy, security, interpretability, and operational drift against its intended use and limits.

REQUIRED OUTCOME

A model-test model with purpose, population, data, labels, baseline, metrics, slices, thresholds, adversarial cases, human review, monitoring, deployment gates, and recovery.

WHY THIS MATTERS

A model can score well overall while failing important populations, contexts, rare events, or safety cases. Test results are sensitive to data, labels, metric choice, threshold, and deployment environment.

APPLICABILITY

Apply when a statistical, machine-learning, generative, rules-based, simulation, or decision model influences output, ranking, prediction, classification, recommendation, or action.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no model behavior, data, decision, generated content, or consequential output changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use model card or purpose, data and privacy, threat, safety, fairness, security, user, deployment, monitoring, and human-review baselines. Define owner, population, threshold, and authority.

REQUIRED INPUTS

Purpose, inputs, outputs, model and data version, labels, baselines, metrics, slices, thresholds, human review, adversarial cases, deployment context, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Model owners define intended use and unacceptable failure; domain and affected-party experts interpret results; humans approve deployment, threshold, exception, high-impact decision, and retirement.

AI RESPONSIBILITIES AND BOUNDARIES

AI may help generate test cases, stratify authorized data, compare results, identify drift, and draft explanations. It must preserve data and model provenance, uncertainty, and privacy; it must not evaluate itself as sole judge, fabricate metrics, infer approval, or deploy or alter a model without authority.

APPLICABLE EXISTING AGENT ROLES

Model Tester, Data Scientist, Researcher, Analyst, Statistician, Privacy Reviewer, Security Auditor, Safety Reviewer, User Experience Reviewer, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define intended and prohibited use; version model and data; choose metrics and slices; test baseline, robustness, fairness, calibration, privacy, security, safety, explainability, and human handoff; analyze error; gate deployment; monitor drift; and retest.

HUMAN-AI INTERACTION REQUIREMENTS

Show model, data, population, slice, metric, threshold, prompt or input context where relevant, result, uncertainty, and limitation. Require human review for high-impact output and use AI judges only with independent controls.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Trace model, data, feature, prompt, tool, policy, provider, threshold, output, reviewer, and action. Support shadow mode, rollback, version comparison, drift detection, human override, privacy controls, and safe degradation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show model and data version, intended use, metrics, slices, drift, incidents, human-review rate, evidence freshness, owner, and authorized pause, rollback, restrict, recalibrate, accept, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Model-test strategy; model and data register; metrics and slice report; robustness, fairness, security, privacy, and safety results; deployment gate; monitoring and recovery; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify model, data, population, slice, label, metric, threshold, environment, method, result, uncertainty, reviewer, and limitation. Aggregate performance is not evidence for every population or use.

VERIFICATION AND ACCEPTANCE TESTS

Verify data lineage, metrics, slices, calibration, robustness, privacy, security, fairness, safety, human review, drift, rollback, and independent reproduction. Test distribution shift and adversarial input.

FAILURE STATES AND ANTI-PATTERNS

Benchmark theater, aggregate-only metrics, leaked test data, self-evaluation, threshold optimized without harm analysis, model drift ignored, and output treated as fact or decision.

RECOVERY AND RECONSIDERATION

Pause or narrow model use, preserve inputs and outputs, route to human or fallback, notify owners, correct affected decisions, and reassess data and model. Reopen on drift, data, model, provider, policy, population, or harm change.

COMPLETION CRITERIA

Purpose, model, data, metrics, slices, limits, human review, evidence, results, recovery, owners, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve model and data versions, test sets, metrics, outputs, decisions, incidents, evidence, exceptions, and open gaps. Hand off to Data, AI Operations, Security, Privacy, Safety, Product, Testing, and Governance.

RELATED CODEX SECTIONS

030-031 Information and Data Architecture; 157-158 AI and Machine Learning Engineering; 217-218 Trust and Privacy; 231 Software Testing; 254 Data Testing; 305 AI Operations; 369-371 Experimentation and Optimization.

CODEX/03_SECTIONS/253_MODEL_TESTING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, AI, data, privacy, safety, and testing context, and the integration roadmap. Exact model, data, population, metrics, fairness, safety, legal, human review, and acceptance authority remain unresolved.

254

SECTION 254 - DATA TESTING

CODEX/03_SECTIONS/254_DATA_TESTING.txt

SHA-256 b7b42e973786373d28ffae41bad2b0d70af02fa4484b30327d0871da573399a9

5,434 BYTES / PRESERVED SOURCE IDENTITY

SECTION 254 — Data Testing

STATUS: CONTENT-COMPLETE

PURPOSE

Verify that data is accurate enough for its declared purpose, complete within scope, consistent, timely, traceable, protected, transformed correctly, and recoverable.

REQUIRED OUTCOME

A data-test model with contracts, schema, grain, quality dimensions, samples, lineage, privacy, reconciliation, freshness, failure handling, evidence, and owner acceptance.

WHY THIS MATTERS

Data can be syntactically valid and still be stale, duplicated, biased, incomplete, misjoined, unauthorized, or semantically wrong. Downstream dashboards or models can amplify a small upstream defect.

APPLICABILITY

Apply when data is collected, transformed, stored, exchanged, reported, modeled, archived, deleted, or used to make or support a decision.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no data meaning, quality, lineage, access, retention, transformation, or consumer behavior changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use data contracts, information and data architecture, privacy, security, quality, warehouse, lake, model, records, and operations baselines. Identify data owner and quality authority.

REQUIRED INPUTS

Source, schema, grain, definitions, population, lineage, transformations, quality rules, thresholds, samples, privacy, access, freshness, reconciliation, consumers, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Data owners define meaning, quality, criticality, access, retention, and acceptance. Domain owners judge semantic correctness; humans approve quarantines, corrections, deletion, and downstream use.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized metadata and bounded samples, draft quality rules, identify anomalies, and propose reconciliation. It must preserve privacy, source, grain, time, and uncertainty and must not expose raw data, change authoritative values, infer sensitive attributes, or declare quality without human review.

APPLICABLE EXISTING AGENT ROLES

Data Tester, Data Engineer, Data Quality Analyst, Data Architect, Privacy Reviewer, Security Auditor, Analyst, Researcher, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define quality and grain; map source and lineage; test schema, completeness, validity, uniqueness, consistency, timeliness, accuracy, reconciliation, privacy, access, transformation, retention, and recovery; quarantine and correct; notify consumers; and retest.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, grain, population, time, rule, sample, result, privacy boundary, owner, and limitation. Require human review for semantic rule, correction, quarantine, deletion, and downstream decision.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support contracts, schema evolution, quality gates, quarantine, lineage, versioned transformations, reconciliation, privacy-preserving fixtures, freshness, rollback, and downstream impact notification.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show data source, grain, quality, freshness, lineage, affected consumers, evidence, incidents, owner, and authorized quarantine, correct, restrict, backfill, restore, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Data-test strategy; contracts and quality rules; source and lineage map; samples and fixtures; results and defect register; quarantine and correction; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, schema, grain, population, time, rule, transformation, sample, method, result, actor, and limitation. Row count, null rate, or passing rule proves only its declared slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify schema, grain, definitions, quality dimensions, lineage, freshness, reconciliation, access, privacy, transformation, correction, backfill, deletion, recovery, and consumer interpretation.

FAILURE STATES AND ANTI-PATTERNS

Quality rule without meaning, row count as accuracy, hidden filters, stale data accepted, sensitive samples copied, quarantine bypassed, and corrected data without lineage or consumer notice.

RECOVERY AND RECONSIDERATION

Quarantine affected data or outputs, preserve versions, notify consumers, correct or backfill, reconcile, restore, and reassess downstream decisions. Reopen on source, schema, definition, transformation, population, or quality change.

COMPLETION CRITERIA

Source, grain, meaning, quality, lineage, privacy, owners, tests, corrections, evidence, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve data and schema versions, rules, samples, results, quarantine, corrections, decisions, evidence, exceptions, and open gaps. Hand off to Data Engineering, Quality, Warehousing, Analytics, AI, Privacy, Security, Testing, and Operations.

RELATED CODEX SECTIONS

030-031 Information and Data Architecture; 153-162 Data Engineering, Quality, Warehousing, Lake, and Analytics; 218 Privacy Engineering; 253 Model Testing; 307 Data Operations; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, data architecture and quality context, and the integration roadmap. Exact definitions, quality thresholds, source authority, privacy, sample, retention, and acceptance evidence remain unresolved.

255

SECTION 255 - TEST AUTOMATION

CODEX/03_SECTIONS/255_TEST_AUTOMATION.txt

SHA-256 4d7ff1c5e5b88ed6a5cd7e8e89301bd7bb125ba2cbf89fcc11afdd0709da6f02

5,043 BYTES / PRESERVED SOURCE IDENTITY

SECTION 255 — Test Automation

STATUS: CONTENT-COMPLETE

PURPOSE

Automate repeatable test execution, data preparation, observation, reporting, and cleanup while preserving truthful scope, maintainability, human judgment, and safe control.

REQUIRED OUTCOME

An automation model with target tests, framework, fixtures, environments, triggers, ownership, flakiness, artifacts, security, observability, maintenance, quarantine, and human exit decision.

WHY THIS MATTERS

Automation increases speed and repeatability but can hide stale tests, weak assertions, data leaks, environment drift, and false confidence. Automation is a tool for evidence, not evidence by itself.

APPLICABILITY

Apply when tests, checks, simulations, scans, or verification steps are executed automatically or are candidates for automation.

NON-APPLICABILITY CONDITIONS

Record non-applicability when automation adds no repeatability or safety value and manual or exploratory judgment is the declared appropriate method.

PREREQUISITES AND DEPENDENCIES

Use test strategy, source control, CI or scheduler, environments, fixtures, secrets, data, observability, artifact storage, access, and maintenance ownership.

REQUIRED INPUTS

Tests, assertions, fixtures, target, framework, trigger, environment, data, dependencies, artifact policy, failure handling, owner, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve automation scope, credentials, data, triggers, quarantine, release gating, and maintenance. Humans interpret exploratory or ambiguous results and retain stop control.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft tests and harnesses, analyze failures, reduce duplication, and propose maintenance. It must not run with unauthorized access, inject uncontrolled data, weaken assertions, hide flakiness, or change release gates without approval.

APPLICABLE EXISTING AGENT ROLES

Test Automation Engineer, Tester, Implementer, Quality Engineer, Security Auditor, Analyst, Verifier, Researcher, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Select stable and valuable cases; design fixtures and isolation; secure credentials; implement and review; execute; capture artifacts; detect flakiness; maintain; quarantine transparently; report coverage and limits; and periodically retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show test, assertion, fixture, environment, credential scope, trigger, result, flakiness, and limitation. Require human approval for generated code, credentials, gate changes, quarantine, and release.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide deterministic reset, artifact identity, secure secret injection, isolated runners, rate limits, cleanup, retry policy, reproducible environments, and clear distinction between test and production authority.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show suite, trigger, target, environment, pass or fail, flakiness, artifacts, owner, evidence freshness, and authorized run, pause, quarantine, repair, accept, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Automation strategy; test and fixture inventory; runner and environment record; artifact and retention policy; flakiness register; results, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify test, code version, fixture, runner, environment, dependency, execution, result, artifact, time, actor, and limitation. Automation completion or coverage proves only the recorded execution.

VERIFICATION AND ACCEPTANCE TESTS

Verify assertions, isolation, credentials, data, cleanup, reproducibility, artifact integrity, retry, flakiness handling, failure visibility, access, and gate behavior.

FAILURE STATES AND ANTI-PATTERNS

Brittle automation, retries hiding failures, secrets in logs, test data leakage, assertions too weak, quarantine without owner, and pipeline green while critical tests are skipped.

RECOVERY AND RECONSIDERATION

Pause unsafe runner, rotate exposed credentials, preserve artifacts, quarantine and repair, rerun, and reassess release evidence. Reopen on target, framework, environment, test, data, or risk change.

COMPLETION CRITERIA

Scope, tests, assertions, fixtures, environment, credentials, artifacts, ownership, maintenance, flakiness, evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve automation code, runner, fixture, environment, artifacts, failures, decisions, exceptions, and open gaps. Hand off to Testing, Quality, Release, Security, Platform, Operations, and Maintenance.

RELATED CODEX SECTIONS

231-254 Testing; 255 Test Automation; 388 Version Management; 301-330 Operations; 386 Technical Debt Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, testing and operations context, and the integration roadmap. Exact framework, runners, data, credentials, artifacts, maintenance, gate, and acceptance authority remain unresolved.

256

SECTION 256 - VERIFICATION

CODEX/03_SECTIONS/256_VERIFICATION.txt

SHA-256 3eb7321c5f1a50afc5552183399f146564cd94b5f3d39790510545b3366e6c70

4,920 BYTES / PRESERVED SOURCE IDENTITY

SECTION 256 — Verification

STATUS: CONTENT-COMPLETE

PURPOSE

Determine, with traceable evidence, whether the specified artifact, process, or system was built and operates according to approved requirements, design, contracts, and constraints.

REQUIRED OUTCOME

A verification record linking requirement or constraint to method, artifact, configuration, test or inspection, result, reviewer, exception, and conclusion within a bounded scope.

WHY THIS MATTERS

Verification answers whether the work conforms to its specification; it does not by itself prove that the specification solves the human problem. Confusing verification with validation creates polished failure.

APPLICABILITY

Apply whenever conformance to requirements, design, contract, standard, configuration, or implementation claim matters.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no conformance claim is made or an approved exact baseline and lower-risk evidence are sufficient for the scope.

PREREQUISITES AND DEPENDENCIES

Use requirements, architecture, contracts, baselines, configuration, tests, inspection, quality, evidence, and acceptance authority. Define independent reviewer where consequence warrants.

REQUIRED INPUTS

Requirement, source, version, artifact, configuration, method, test, inspection, evidence, result, exception, reviewer, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define conformance and exceptions; verifiers assess evidence; humans decide material deviation, waiver, and acceptance. The verifier must not claim more than the method demonstrates.

AI RESPONSIBILITIES AND BOUNDARIES

AI may trace requirements, inspect authorized artifacts, draft checks, and summarize results. It must not fabricate execution, infer conformance from similarity, alter evidence, or sign verification without human review.

APPLICABLE EXISTING AGENT ROLES

Verifier, Tester, Quality Assurance Reviewer, Requirements Engineer, Architect, Analyst, Researcher, Critic, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Define conformance; baseline sources and versions; select method; inspect, analyze, or test; record evidence and deviations; review independently; issue bounded conclusion; track correction; and reopen on change.

HUMAN-AI INTERACTION REQUIREMENTS

Show requirement, source, artifact, version, method, evidence, result, exception, reviewer, and limitation. Require human judgment for semantic interpretation, waiver, and conclusion.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Maintain traceability, immutable baselines, configuration identity, test reproducibility, independent evidence, and change impact. Keep verification records separate from unverified claims.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show requirement, conformance state, evidence freshness, deviation, owner, reviewer, and authorized verify, reject, waive, remediate, reopen, and accept controls.

REQUIRED OUTPUTS AND ARTIFACTS

Verification plan; traceability matrix; method and evidence record; results and deviations; reviewer conclusion; remediation, exception, decision, and continuity records.

EVIDENCE REQUIREMENTS

Identify requirement, version, artifact, environment, method, execution or inspection, result, reviewer, time, and limitation. A test result proves only the tested requirement and conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify source authority, version, traceability, method validity, evidence integrity, result reproducibility, deviation handling, independence, and conclusion boundary.

FAILURE STATES AND ANTI-PATTERNS

Verification by assertion, stale requirement, untraceable test, conformance inferred from intent, waiver without authority, and verification presented as user validation.

RECOVERY AND RECONSIDERATION

Mark conclusion uncertain, preserve evidence, correct artifact or requirement trace, obtain review, and reverify. Reopen on requirement, design, implementation, configuration, environment, or evidence change.

COMPLETION CRITERIA

Requirement, artifact, method, evidence, result, deviations, reviewer, conclusion, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending independent review and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve baselines, traceability, evidence, results, deviations, decisions, exceptions, and open gaps. Hand off to Validation, Acceptance, Quality, Release, Operations, Audit, and Governance.

RELATED CODEX SECTIONS

004 Evidence and Truth; 019 Requirements; 228-230 Quality; 231-255 Testing; 257 Validation; 237 Acceptance Testing; 402-404 Review.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, requirements and evidence context, and the integration roadmap. Exact conformance criteria, independence, methods, standards, deviations, and acceptance authority remain unresolved.

257

SECTION 257 - VALIDATION

CODEX/03_SECTIONS/257_VALIDATION.txt

SHA-256 98102f63d04270e3128eb0688bc3718db1f9463f9a716430ce94956493069d51

5,403 BYTES / PRESERVED SOURCE IDENTITY

SECTION 257 — Validation

STATUS: CONTENT-COMPLETE

PURPOSE

Determine whether the built capability solves the intended problem, supports real users and owners, produces acceptable outcomes, and remains suitable in its declared context.

REQUIRED OUTCOME

A validation record connecting human objective, users, context, scenarios, outcomes, evidence, alternatives, harms, acceptance, and unresolved mismatch to a decision owner.

WHY THIS MATTERS

A system can conform perfectly to requirements that were wrong, incomplete, inaccessible, or detached from lived reality. Validation requires evidence from the people and conditions that give the work meaning.

APPLICABILITY

Apply when a capability, product, service, policy, model, process, or decision must be shown useful, safe, acceptable, or fit for its intended purpose.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no intended outcome, user, stakeholder, operational purpose, or material consequence is affected and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use research, requirements, user journeys, domain, safety, privacy, accessibility, quality, verification, acceptance, operations, and outcome evidence. Identify validation owner and affected parties.

REQUIRED INPUTS

Objective, users, affected non-users, context, scenarios, desired outcomes, requirements, artifact, evidence, alternatives, risks, harms, feedback, and acceptance authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans and representative users judge usefulness, fit, acceptability, harm, and tradeoffs. Owners decide whether evidence supports adoption; affected people must be able to report mismatch and withdraw or appeal where applicable.

AI RESPONSIBILITIES AND BOUNDARIES

AI may synthesize authorized research, compare outcomes, draft validation scenarios, and surface mismatch or uncertainty. It must not fabricate user evidence, speak for users, infer acceptance, or declare fitness from conformance or proxy metrics.

APPLICABLE EXISTING AGENT ROLES

Validator, User Researcher, User Experience Reviewer, Researcher, Analyst, Domain Analyst, Safety Reviewer, Accessibility Reviewer, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Restate intended outcome; identify users and context; test realistic scenarios; observe benefit, harm, comprehension, and recovery; compare alternatives; collect dissent; assess unintended effects; decide; monitor after use; and reopen when reality changes.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, user, context, scenario, evidence, outcome, uncertainty, harm, alternative, and decision authority. Require direct human or user evidence where material and never present AI synthesis as participation.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support user feedback, outcome instrumentation, accessibility, explainability, appeal, safety, privacy, rollback, support, and post-deployment learning. Separate proxy metrics from actual human outcomes.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show intended outcome, users, evidence, observed benefit and harm, dissent, owner, acceptance, limitations, and authorized pause, revise, compensate, reject, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Validation charter; user and context map; scenario and outcome record; evidence and dissent ledger; harm and alternative assessment; decision, acceptance, monitoring, and continuity records.

EVIDENCE REQUIREMENTS

Identify user or stakeholder, objective, context, scenario, method, time, observation, outcome, impact, uncertainty, and limitation. Conformance, satisfaction, usage, or model score alone does not prove fitness.

VERIFICATION AND ACCEPTANCE TESTS

Verify representative users and context, intended outcome, actual use, accessibility, safety, privacy, support, harm, alternatives, dissent, and post-use monitoring. Reproduce material findings and document unresolved mismatch.

FAILURE STATES AND ANTI-PATTERNS

Validation by proxy metric, requirement compliance treated as usefulness, ideal-user testing, dissent erased, harm externalized, and adoption declared from launch or usage alone.

RECOVERY AND RECONSIDERATION

Pause or narrow use, protect affected people, preserve evidence, remedy harm, revise design or objective, and gather missing perspective. Reopen on user, context, outcome, population, policy, technology, or harm change.

COMPLETION CRITERIA

Objective, users, context, scenarios, outcomes, evidence, harms, alternatives, decision, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending affected-party and owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve research, validation, user evidence, dissent, decisions, acceptance, monitoring, incidents, compensation, and open gaps. Hand off to Product, Requirements, UX, Quality, Operations, Support, Governance, and Review.

RELATED CODEX SECTIONS

001 User Research; 019 Requirements; 237 Acceptance Testing; 249 Usability Testing; 256 Verification; 301-330 Operations; 402-404 Review and Lessons Learned.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, research, requirements, user, quality, and evidence context, and the integration roadmap. Exact users, outcomes, context, harm, evidence, participation, and acceptance authority remain unresolved.